

Digital System Design

LAB I

Vivado and Digilent Board

DONG Yunyang

dongyy@sustech.edu.cn

411, No. 2, Hui Yuan

Tencent Meeting: 874-068-9694

Course Arrangement

- Part 1: Lecture
- Part 2: Lab
 - VHDL / FPGA board / Vivado
 - Exercise (Demo & Report)
 - Group Project
- TA: 曹国群 caogq2024@mail.sustech.edu.cn
- 吴浩源 12432640@mail.sustech.edu.cn
- 刘备尊 liubz2024@mail.sustech.edu.cn

Digital System Hardware

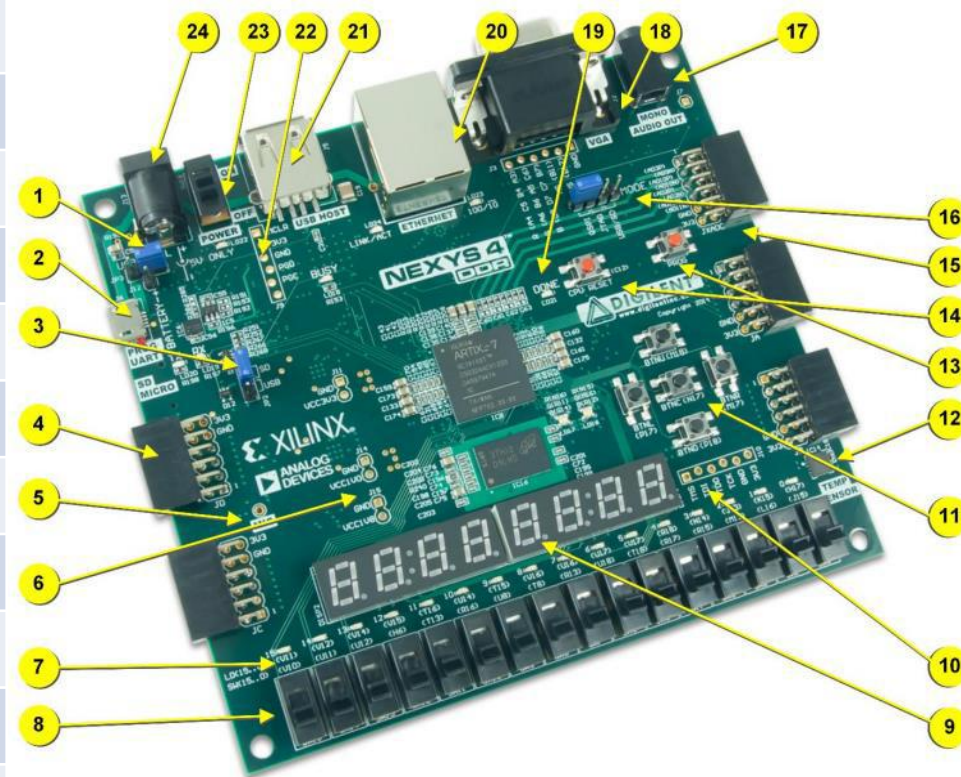
- 标准芯片
 - 74LS00、74LS138、74LS160
 - 真值表
 - 卡诺图
- 可编程逻辑器件(Programmable Logic Devices)
 - FPGA(Field-Programmable Gate Array)
 - 硬件描述语言
 - VHDL
 - Verilog HDL
- 自定义芯片



FPGA Development Board

- Nexys A7 板（又名Nexys4 DDR）是Digilent公司设计的一款FPGA板卡。
- 板卡采用了Xilinx Artix-7 100T FPGA芯片。

1	供电跳线	13	FPGA配置复位键
2	UART/JTAG共用USB	14	CPU复位键
3	外部配置跳线(SD/USB)	15	模拟信号Pmod(XADC)
4	Pmod端口	16	编程模式跳线
5	传声器	17	音频接口
6	电源测试点	18	VGA接口
7	LED	19	编程完成LED
8	拨动开关	20	以太网接口
9	7段数码管	21	USB接口
10	外部接线JTAG端口	22	PIC24编程端口
11	按键开关	23	电源开关
12	温度传感器	24	电源接口



EDA Tool

对于Xilinx芯片采用配套的开发软件：Vivado

Start > All Programs > Xilinx Design Tools > Vivado 2020.2

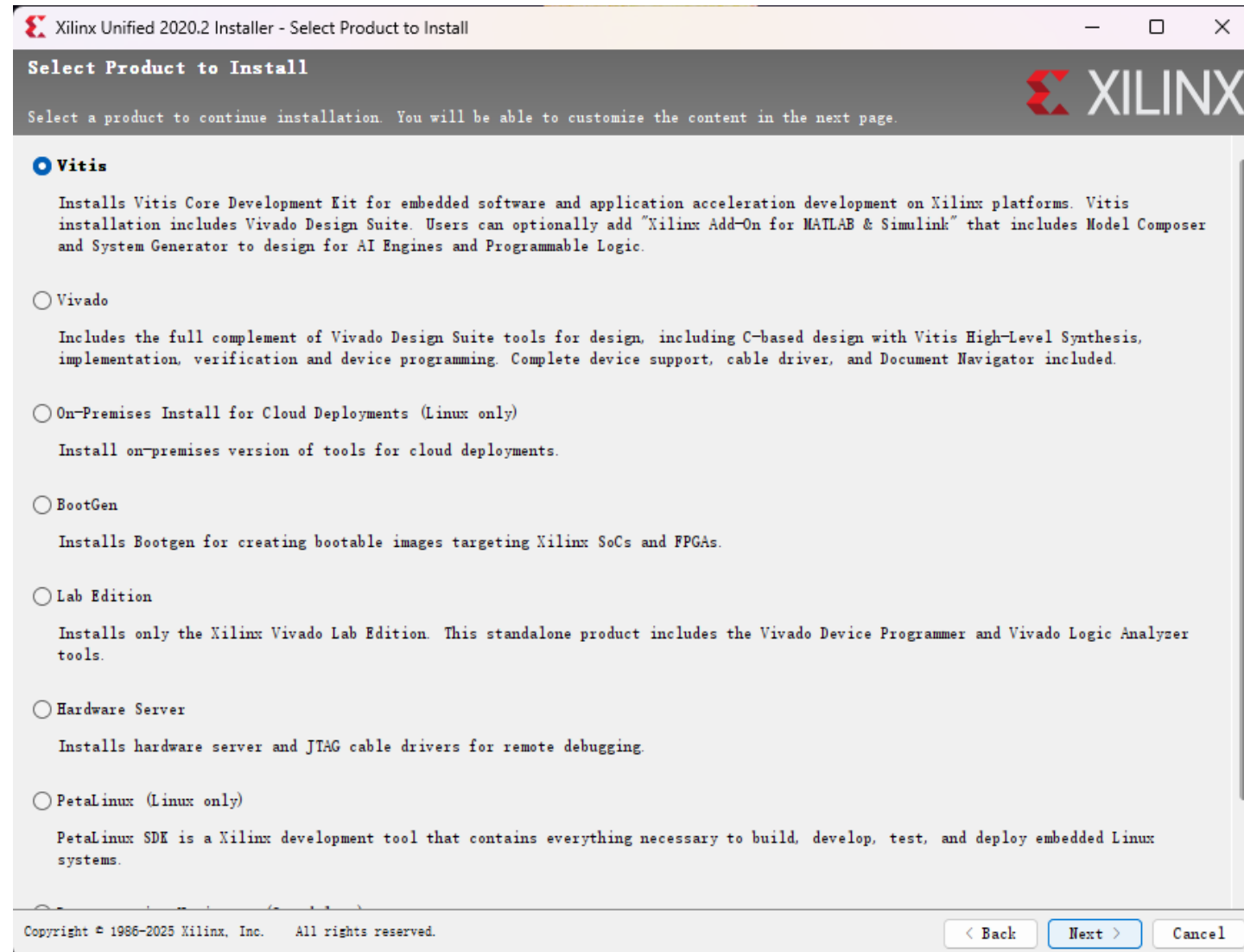
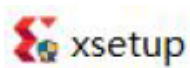


安装：

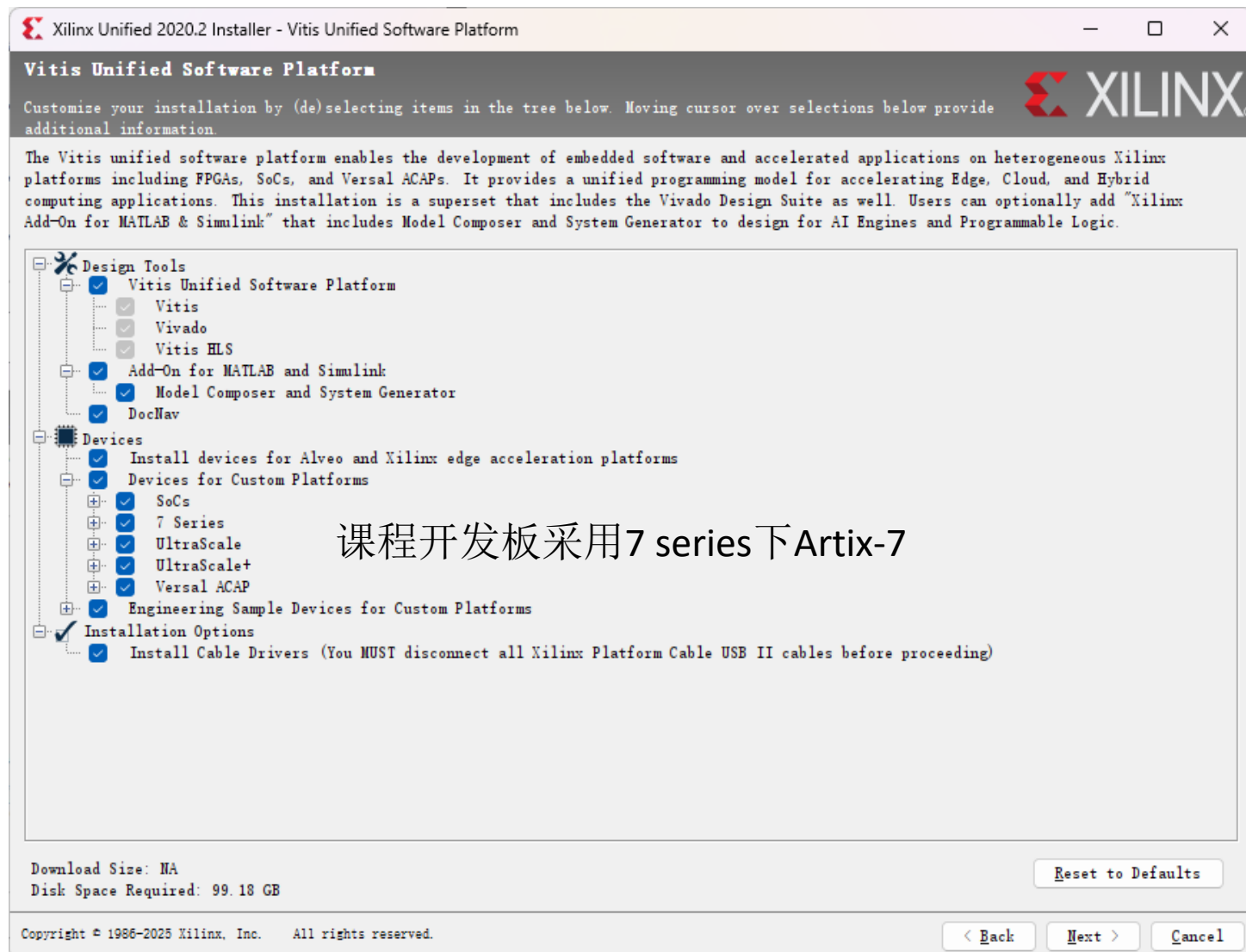
<https://www.xilinx.com/support/download.html>

安装后占用磁盘空间非常大，请预留100GB以上的空间

Vivado Installation



Vivado Installation



课程开发板采用7 series下Artix-7

Vivado Installation

Xilinx Unified 2020.2 Installer - Accept License Agreements

Accept License Agreements

Please read the following terms and conditions and indicate that you agree by checking the I Agree checkboxes.

Xilinx Inc. End User License Agreement

By checking "I Agree" below, or OTHERWISE ACCESSING, DOWNLOADING, INSTALLING or USING THE SOFTWARE, I AGREE on behalf of licensee to be bound by the agreement, which can be viewed by [clicking here](#).

☒ **I Agree**

WebTalk Terms And Conditions

By checking "I Agree" below, I also confirm that I have read [Section 13 of the terms and conditions](#) above concerning WebTalk and have been afforded the opportunity to read the WebTalk FAQ posted at <https://www.xilinx.com/products/design-tools/webtalk.html>. I understand that I am able to disable WebTalk later if certain criteria described in Section 13(c) apply. If they don't apply, I can disable WebTalk by uninstalling the Software or using the Software on a machine not connected to the internet. If I fail to satisfy the applicable criteria or if I fail to take the applicable steps to prevent such transmission of information, I agree to allow Xilinx to collect the information described in Section 13(a) for the purposes described in Section 13(b).

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☒ **I Agree**

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< Back Next > Cancel

Vivado Installation

Xilinx Unified 2020.2 Installer - Select Destination Directory

Select Destination Directory

Choose installation options such as location and shortcuts.

Installation Options

Select the installation directory

D:\Xilinx

Installation location(s)

D:\Xilinx\Vitis\2020.2
D:\Xilinx\Vivado\2020.2
D:\Xilinx\Vitis_HLS\2020.2
D:\Xilinx\Model_Composer\2020.2
D:\Xilinx\DocNav

Disk Space Required

Download Size:	NA
Disk Space Required:	99.18 GB
Final Disk Usage:	67.57 GB
Disk Space Available:	233.02 GB

Select shortcut and file association options

☒ Create program group entries

Xilinx Design Tools

☒ Create desktop shortcuts

☒ Create file associations

Apply shortcut & file association selections to

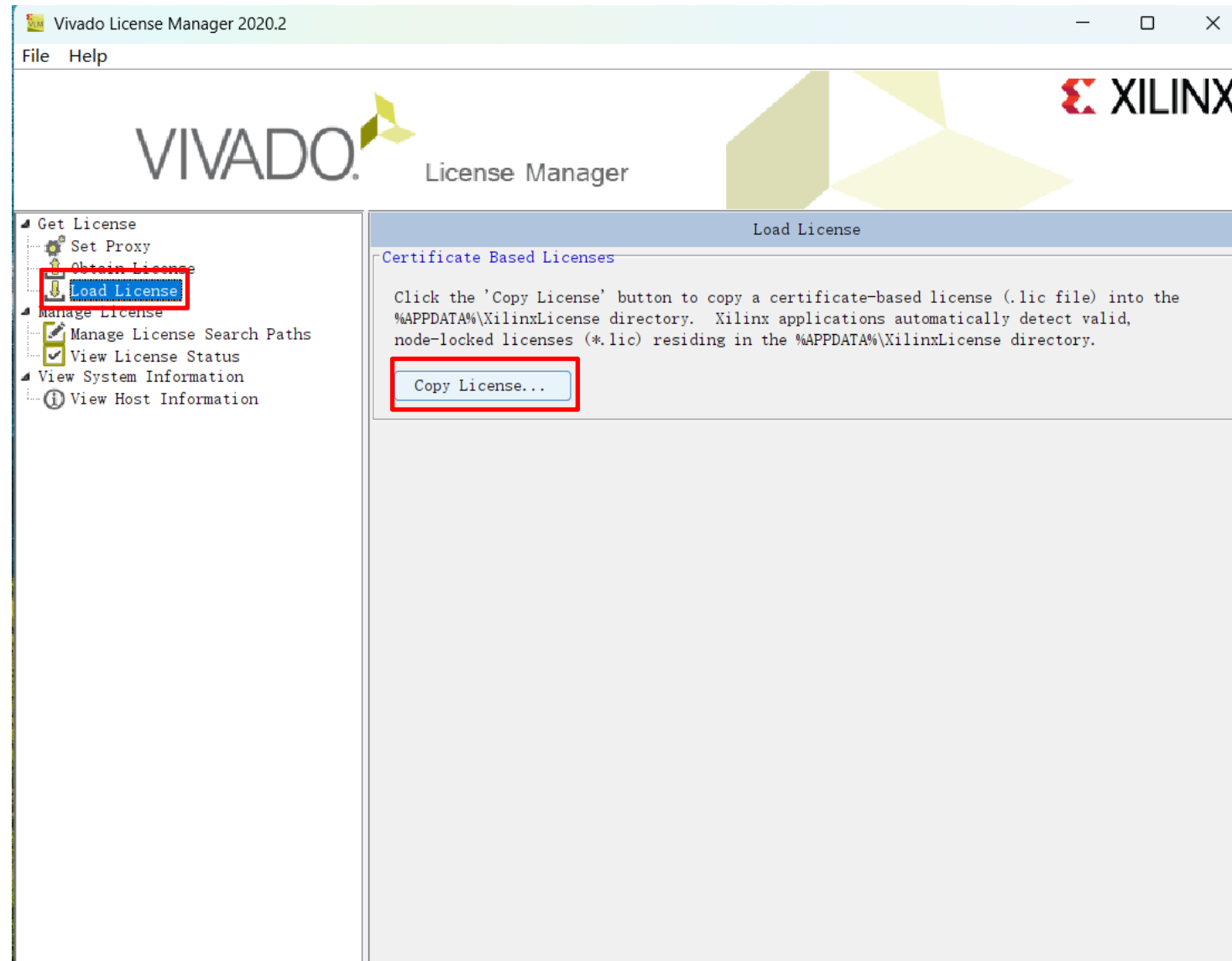
☐ Current user

☒ All users

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< Back Next > Cancel

Vivado Installation

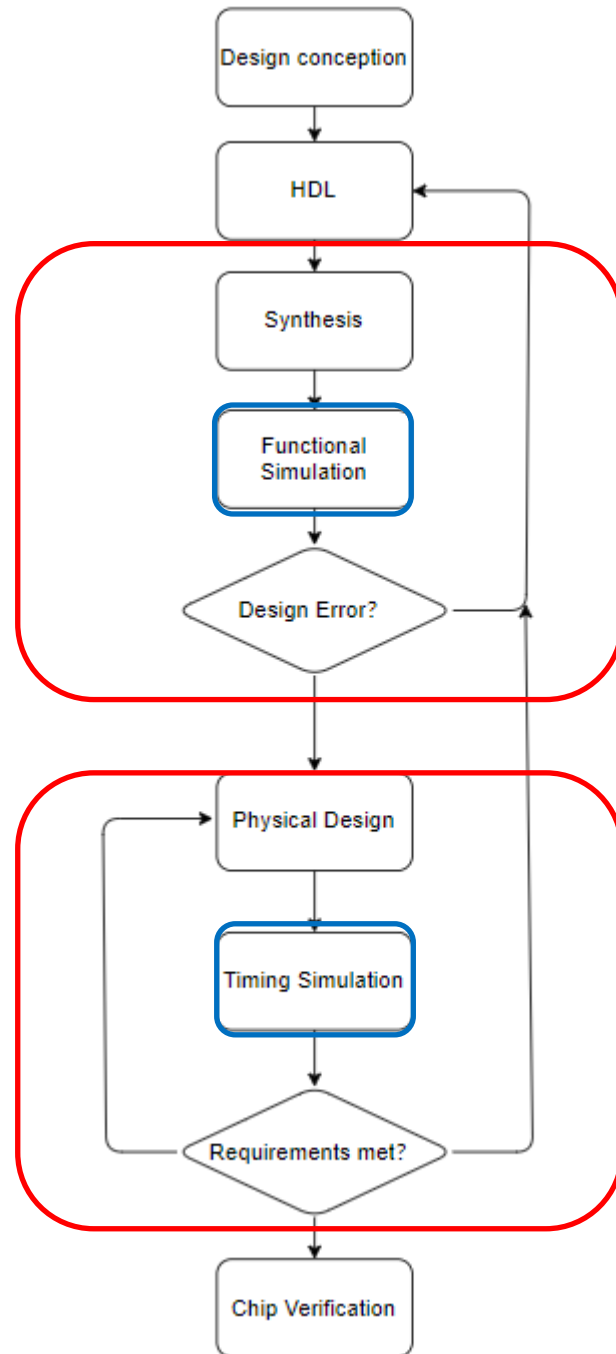


EDA Flowchart

RTL Analysis
Synthesis

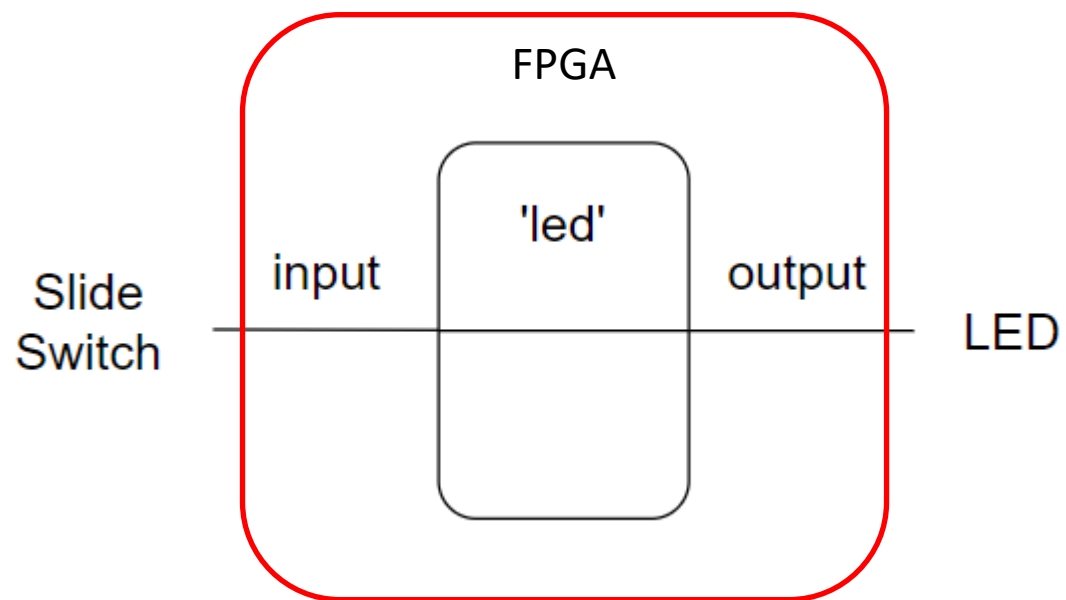
Implementation

Program and Debug



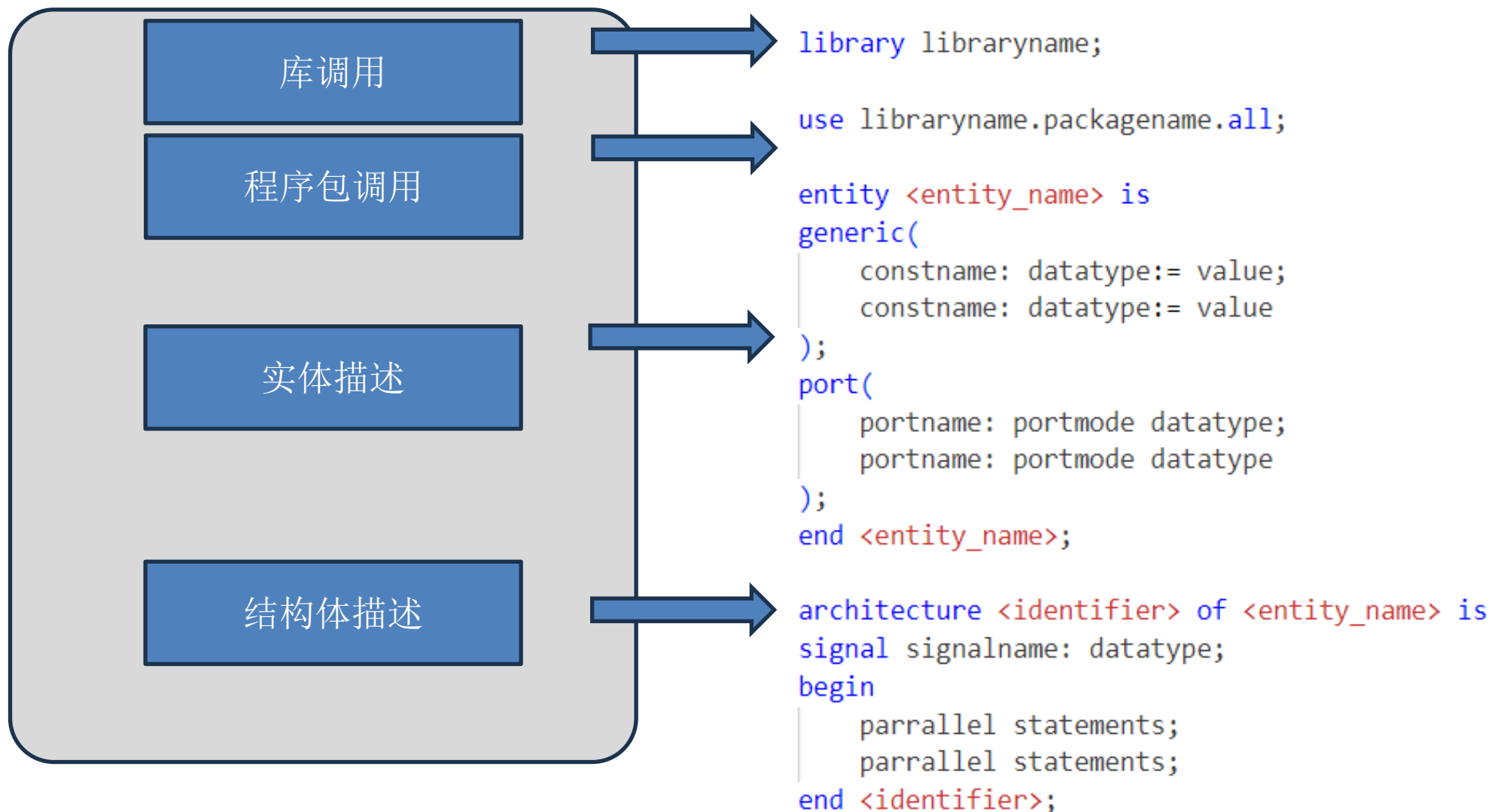
The First Design

设计如图所示的系统

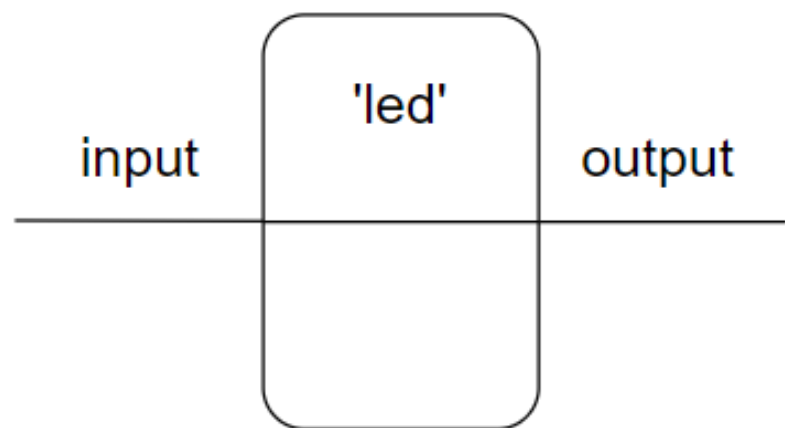


VHDL

VHDL基本结构

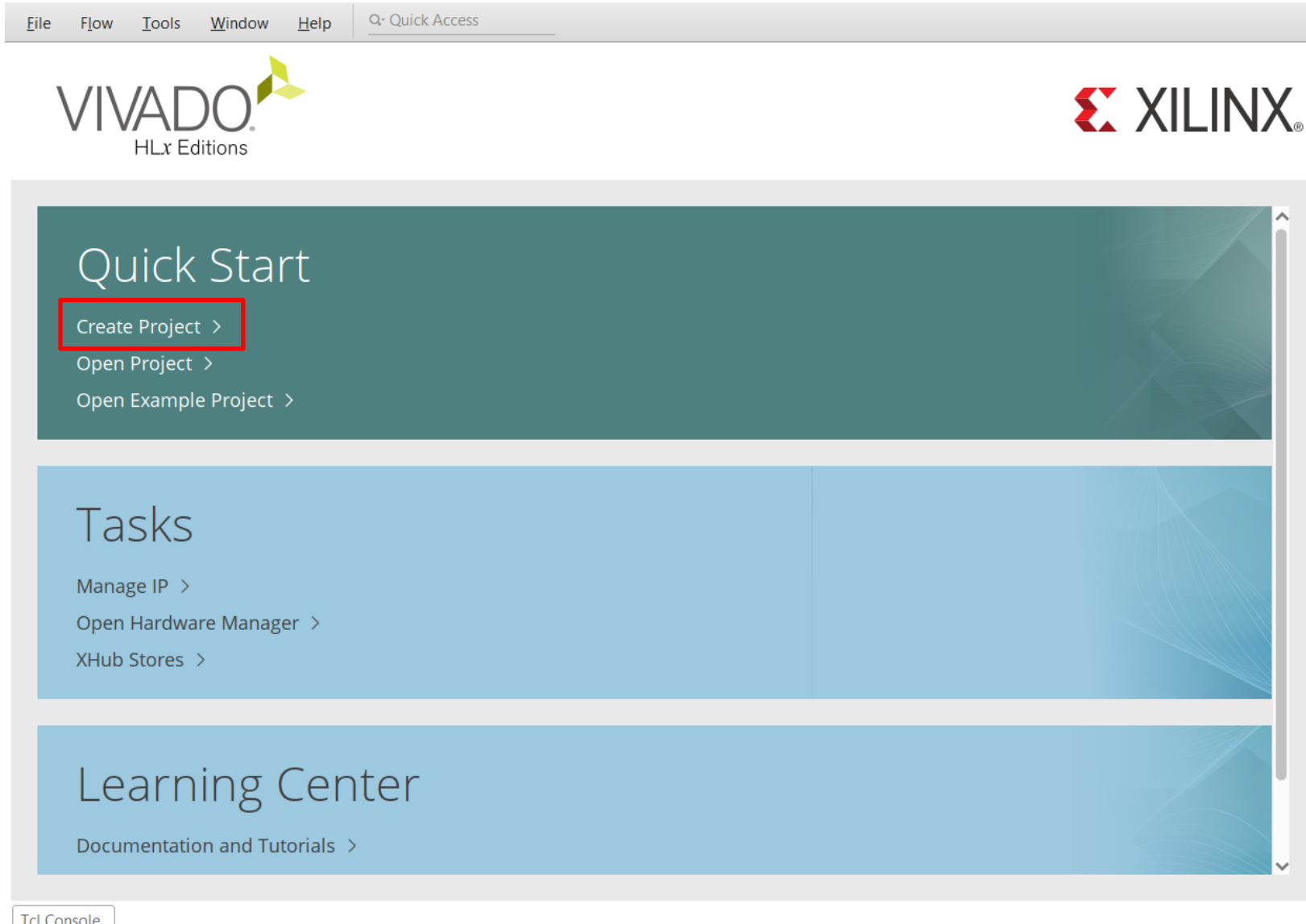


VHDL



```
library ieee;  
use ieee.std_logic_1164.all;  
  
entity led is  
port(  
    input    :in    std_logic;  
    output   :out    std_logic  
);  
end led;  
  
architecture behavior of led is  
begin  
    output <= input;  
end behavior;
```

Create Project



Create Project

Project Name

Enter a name for your project and specify a directory where the project data files will be stored.

Project name: Tutorial

Project location: F:/FPGA

☒ Create project subdirectory

Project will be created at: F:/FPGA/Tutorial

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< Back

Next >

Finish

Cancel

Create Project

Project Type

Specify the type of project to create.



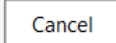
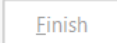
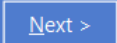
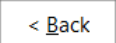
☒ **RTL Project**
You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.
☐ Do not specify sources at this time
☐ Project is an extensible Vitis platform

☐ **Post-synthesis Project**
You will be able to add sources, view device resources, run design analysis, planning and implementation.
☐ Do not specify sources at this time

☐ **I/O Planning Project**
Do not specify design sources. You will be able to view part/package resources.

☐ **Imported Project**
Create a Vivado project from a Synplify, XST or ISE Project File.

☐ **Example Project**
Create a new Vivado project from a predefined template.



Create Project

Add Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.



Use Add Files, Add Directories or Create File buttons below

Add Files

Add Directories

Create File

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

Target language: VHDL

Simulator language: Mixed



< Back

Next >

Finish

Cancel

Create Project

Add Constraints (optional)
Specify or create constraint files for physical and timing constraints.

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Use Add Files or Create File buttons below

Add Files

Create File

☐ Copy constraints files into project

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< Back

Next >

Finish

Cancel

Create Project

芯片选型

Default Part

Choose a default Xilinx part or board for your project.



Parts Boards

[Reset All Filters](#)

Category: All
Family: Artix-7

Package: csg324
Speed: -1

Temperature: All Remaining
Static power: All Remaining

Search: Q

Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs	Gb Transceivers	GTPE2 Transceivers
xc7a15tcsg324-1	324	210	10400	20800	25	0	45	0	0
xc7a35tcsg324-1	324	210	20800	41600	50	0	90	0	0
xc7a50tcsg324-1	324	210	32600	65200	75	0	120	0	0
xc7a75tcsg324-1	324	210	47200	94400	105	0	180	0	0
xc7a100tcsg324-1	324	210	63400	126800	135	0	240	0	0



< Back

Next >

Finish

Cancel

Create Project

或者

Default Part
Choose a default Xilinx part or board for your project.

Parts

Boards

[Reset All Filters](#)

Install/Update Boards

Vendor: digilentinc.com

Name: Nexys4 DDR

Board Rev: Latest

Search: Q:

Display Name	Preview	Vendor	File Version	Part	I/O Pin Count
Nexys4 DDR		digilentinc.com	1.1	xc7a100tcsg324-1	324

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Next >

Finish

Cancel

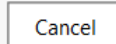
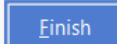
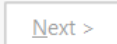
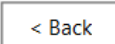
Create Project



New Project Summary

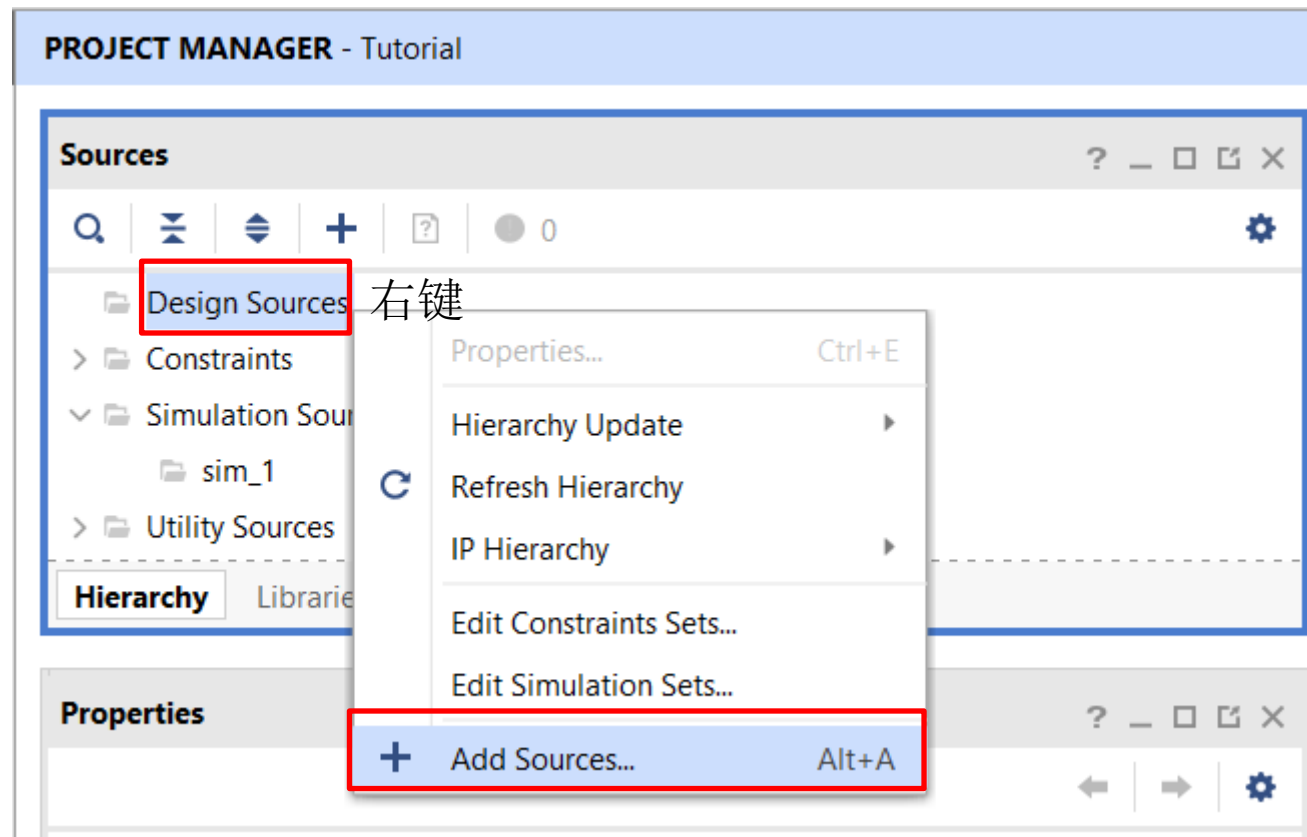
-  A new RTL project named 'Tutorial' will be created.
-  No source files or directories will be added. Use Add Sources to add them later.
-  No constraints files will be added. Use Add Sources to add them later.
-  The default part and product family for the new project:
 - Default Part: xc7a100tcsg324-1
 - Product: Artix-7
 - Family: Artix-7
 - Package: csg324
 - Speed Grade: -1

To create the project, click Finish



Create VHDL Source File

添加设计文件



Create VHDL Source File



Add Sources

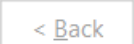
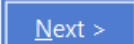
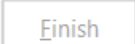
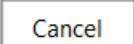
This guides you through the process of adding and creating sources for your project

☐ Add or create constraints

☒ Add or create design sources

☐ Add or create simulation sources



Create VHDL Source File

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

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Use Add Files, Add Directories or Create File buttons below

Add Files

Add Directories

Create File

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

?

< Back

Next >

Finish

Cancel

Create VHDL Source File

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

Create Source File

Create a new source file and add it to your project.

File type:

VHDL

File name:

led

File location:

<Local to Project>

?

OK

Cancel

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

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< Back

Next >

Finish

Cancel

Create VHDL Source File

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

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	Index	Name	Library	Location
	1	led.vhd	xil_defaultlib	<Local to Project>

Add Files

Add Directories

Create File

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

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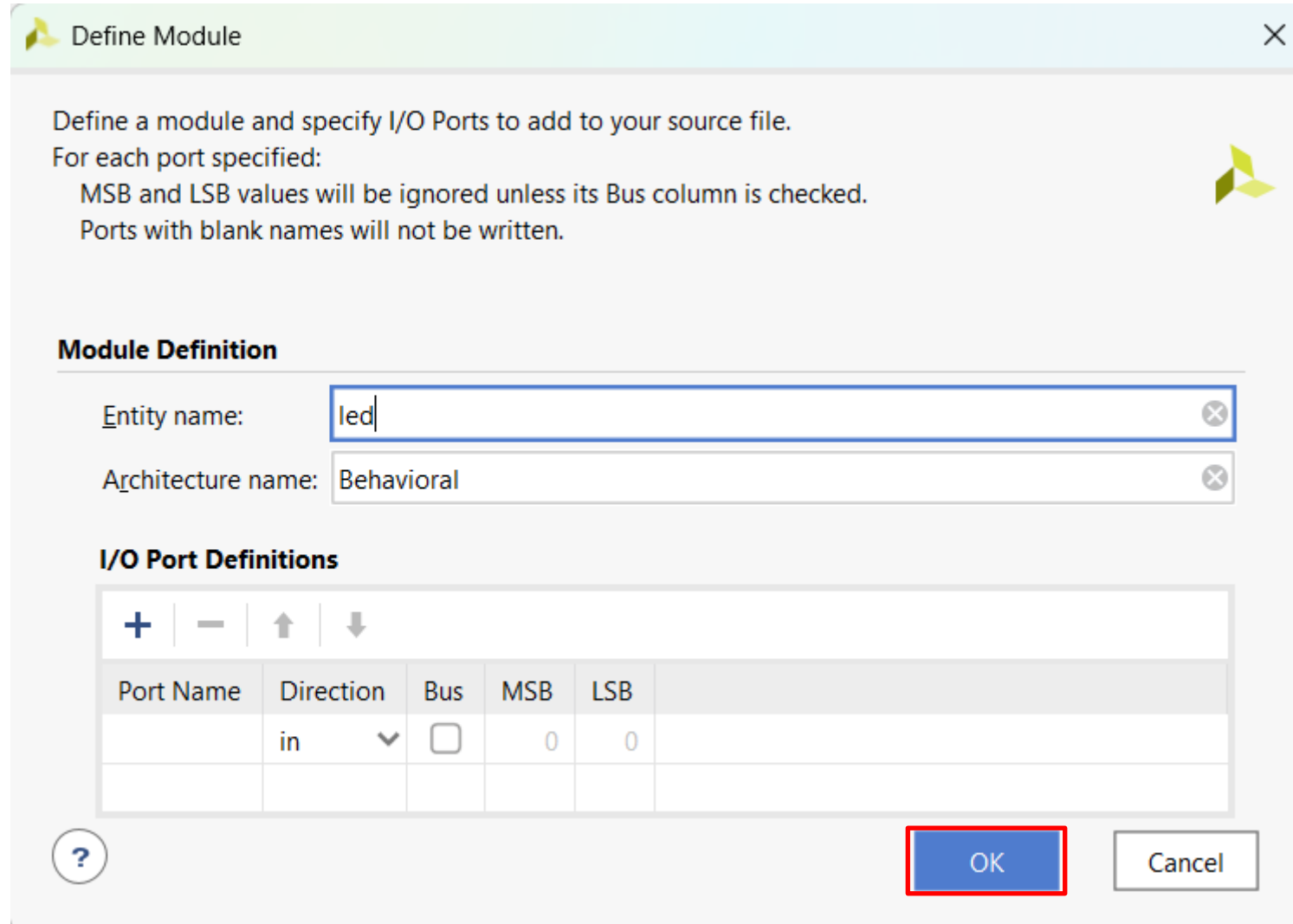
< Back

Next >

Finish

Cancel

Create VHDL Source File

A screenshot of the 'Define Module' dialog box in a VHDL IDE. The dialog has a title bar with a yellow cube icon and a close button. The main area contains instructions: 'Define a module and specify I/O Ports to add to your source file. For each port specified: MSB and LSB values will be ignored unless its Bus column is checked. Ports with blank names will not be written.' Below this is the 'Module Definition' section with two text boxes: 'Entity name:' containing 'led' and 'Architecture name:' containing 'Behavioral'. The 'I/O Port Definitions' section features a table with columns: Port Name, Direction, Bus, MSB, and LSB. The first row has 'in' in the Direction column, an unchecked checkbox in the Bus column, and '0' in both MSB and LSB columns. Above the table are icons for adding (+), removing (-), and moving up/down. At the bottom are a help button (question mark in a circle), an 'OK' button (highlighted with a red rectangle), and a 'Cancel' button.

Define Module

Define a module and specify I/O Ports to add to your source file.
For each port specified:
MSB and LSB values will be ignored unless its Bus column is checked.
Ports with blank names will not be written.

Module Definition

Entity name: led

Architecture name: Behavioral

I/O Port Definitions

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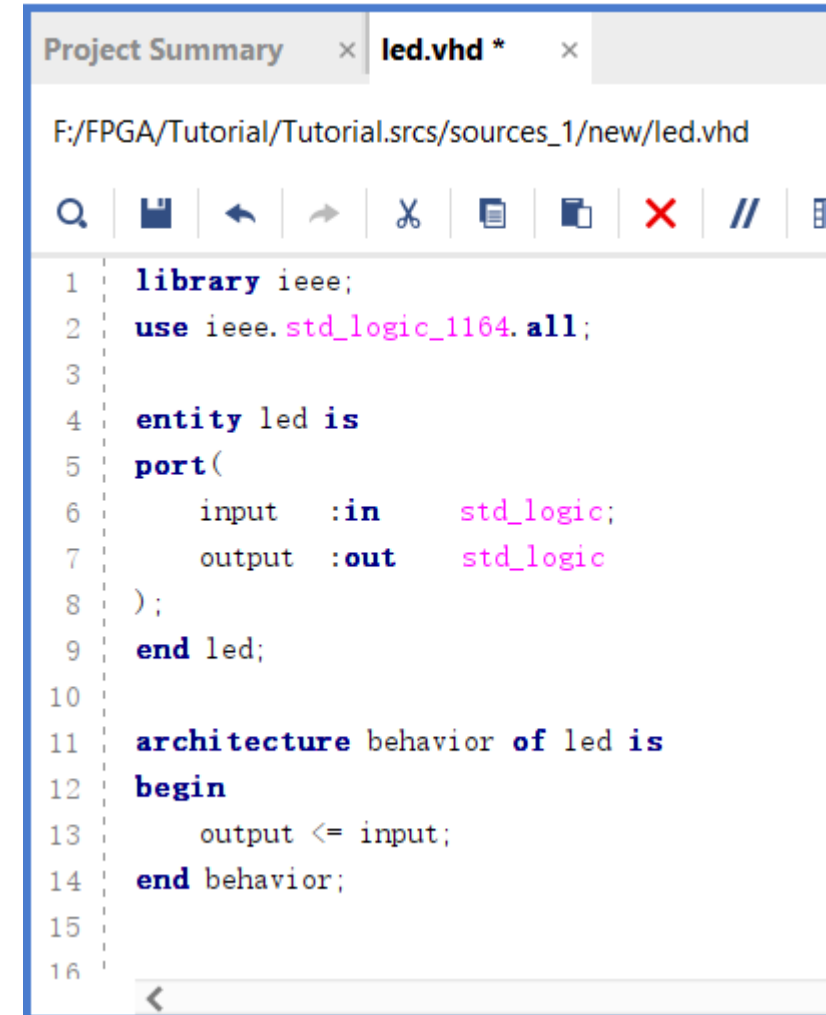
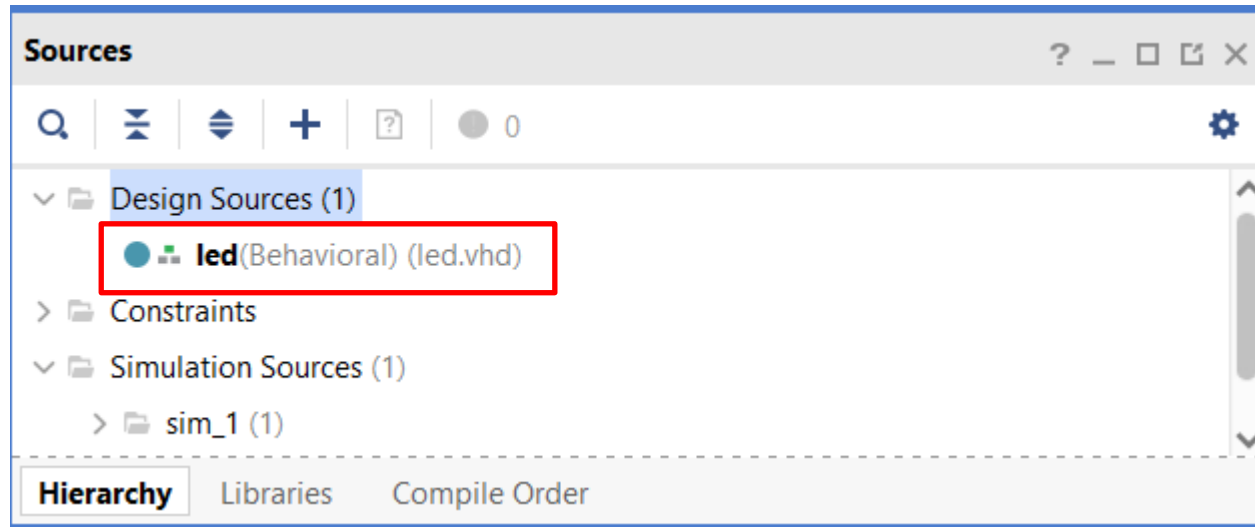
Port Name	Direction	Bus	MSB	LSB
	in	☐	0	0

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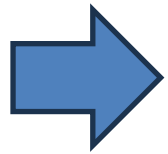
OK

Cancel

Create VHDL Source File



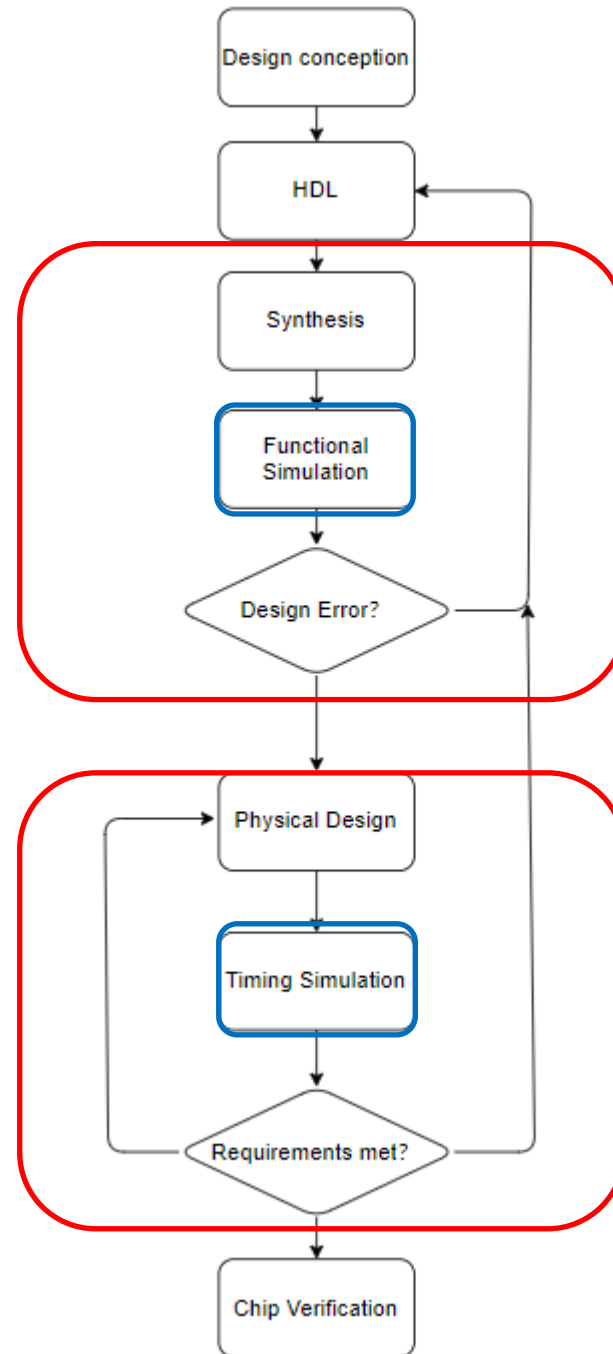
EDA Flowchart



RTL Analysis
Synthesis

Implementation

Program and Debug



RTL Analysis

▼ PROJECT MANAGER

⚙ Settings

Add Sources

Language Templates

🔗 IP Catalog

▼ IP INTEGRATOR

Create Block Design

Open Block Design

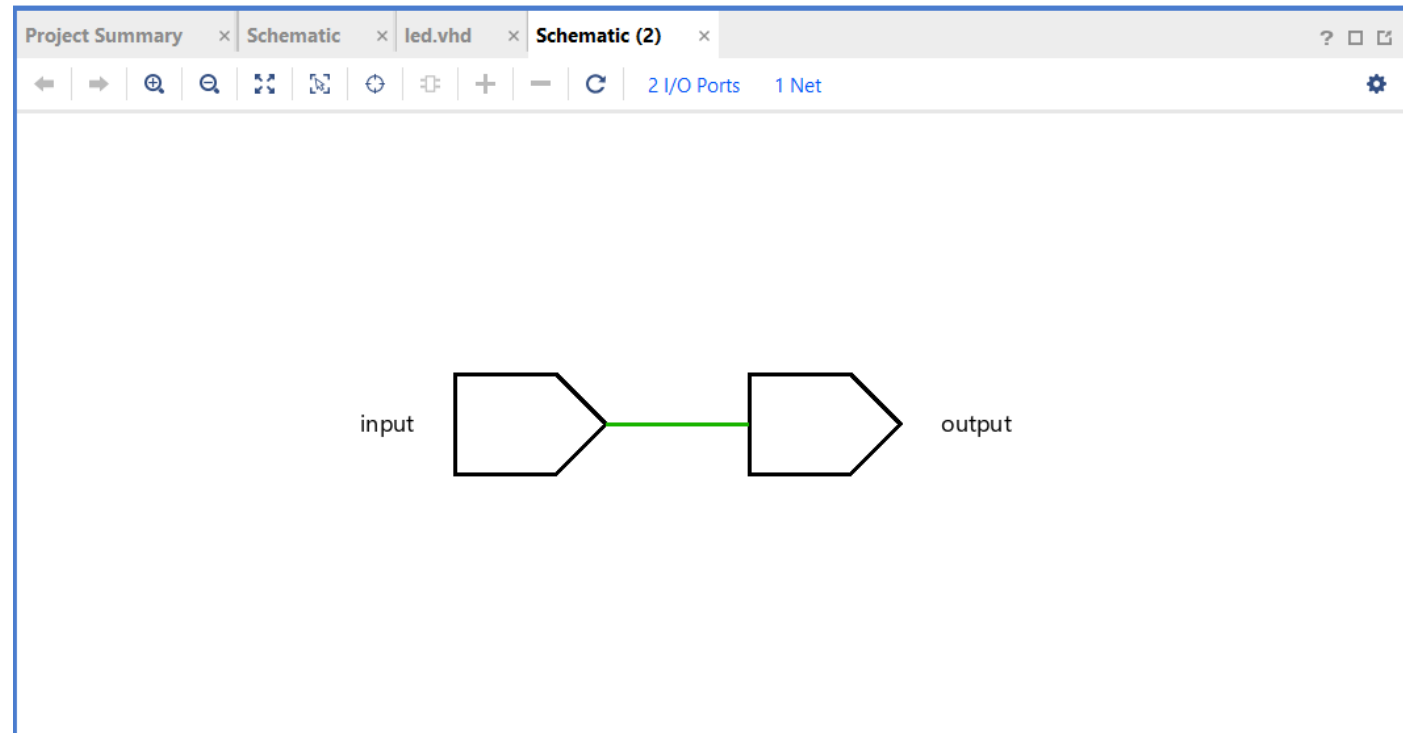
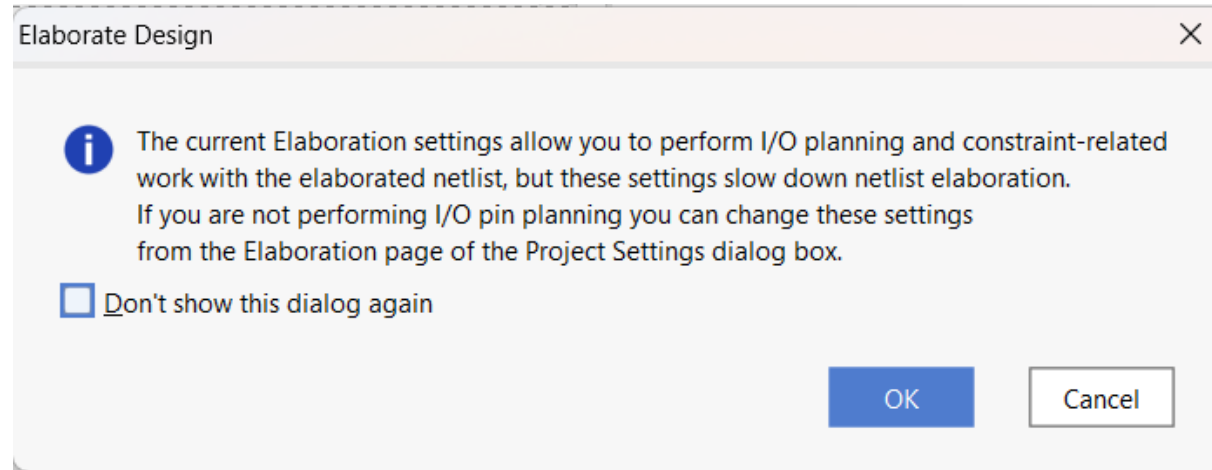
Generate Block Design

▼ SIMULATION

Run Simulation

▼ RTL ANALYSIS

> Open Elaborated Design



Synthesize the Design

▼ SIMULATION

Run Simulation

▼ RTL ANALYSIS

▼ Open Elaborated Design

☒ Report Methodology

Report DRC

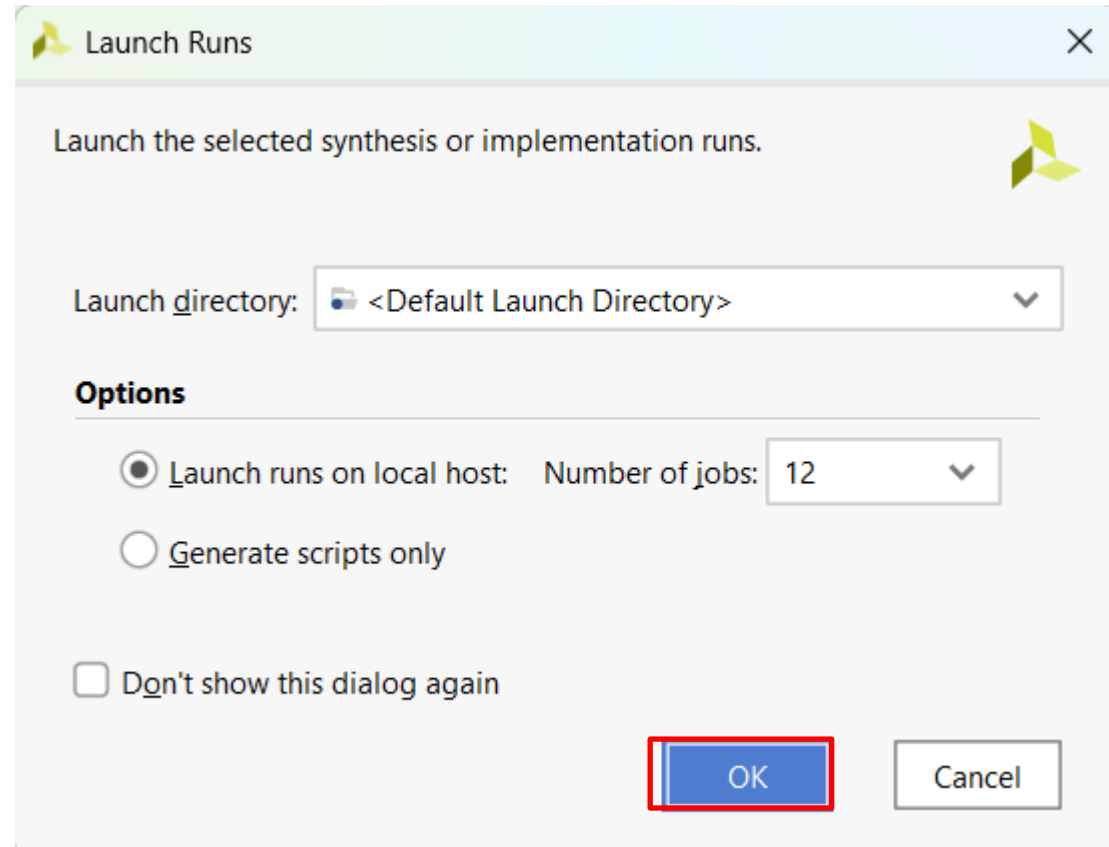
Report Noise

 Schematic

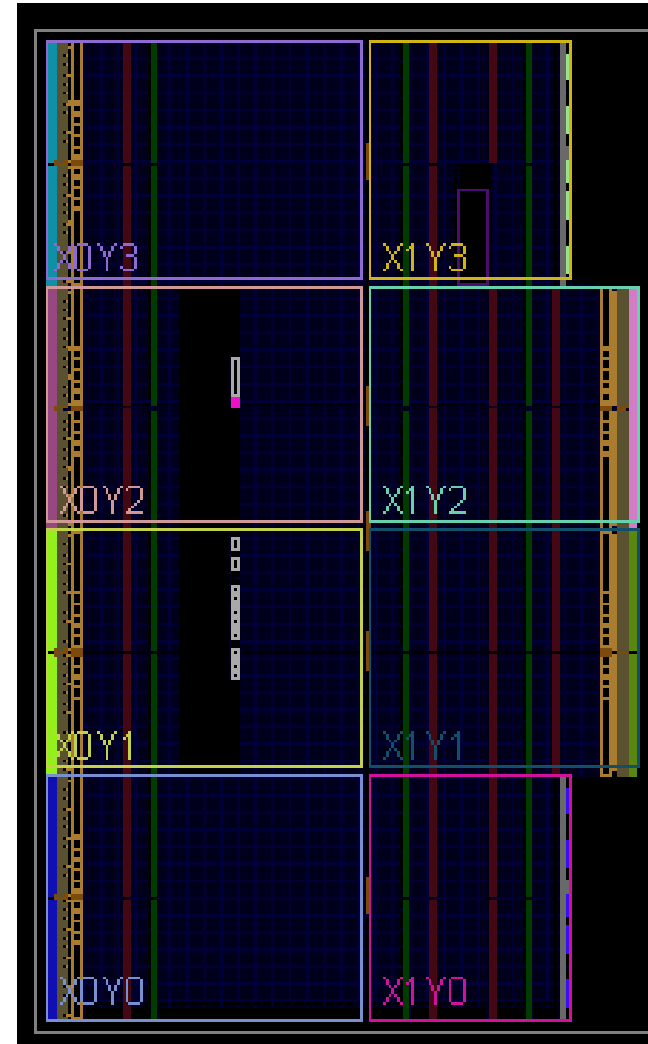
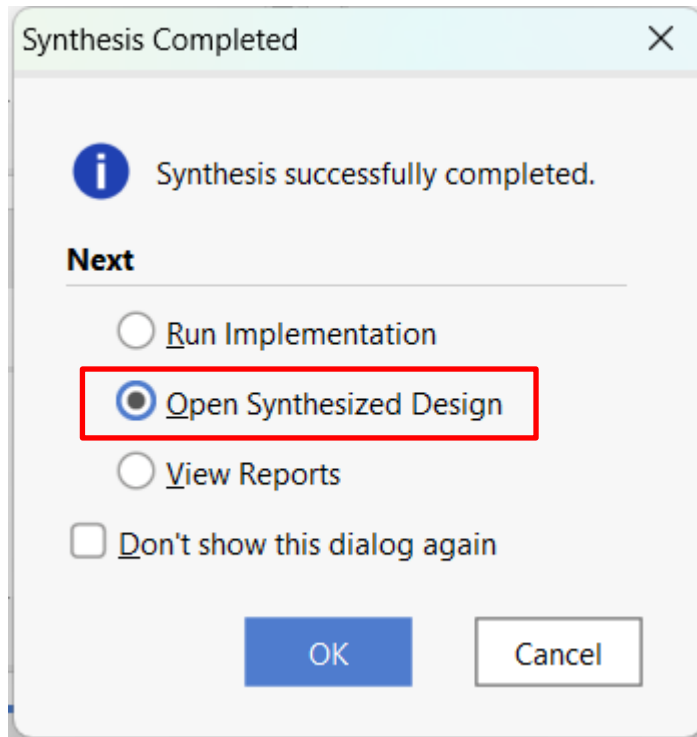
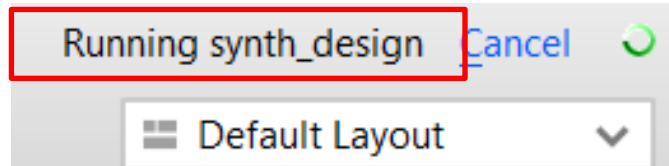
▼ SYNTHESIS

 Run Synthesis

> Open Synthesized Design



Synthesize the Design



Synthesize the Design

SYNTHESIS

Run Synthesis

Open Synthesized Design

Constraints Wizard

Edit Timing Constraints

Set Up Debug

Report Timing Summary

Report Clock Networks

Report Clock Interaction

Report Methodology

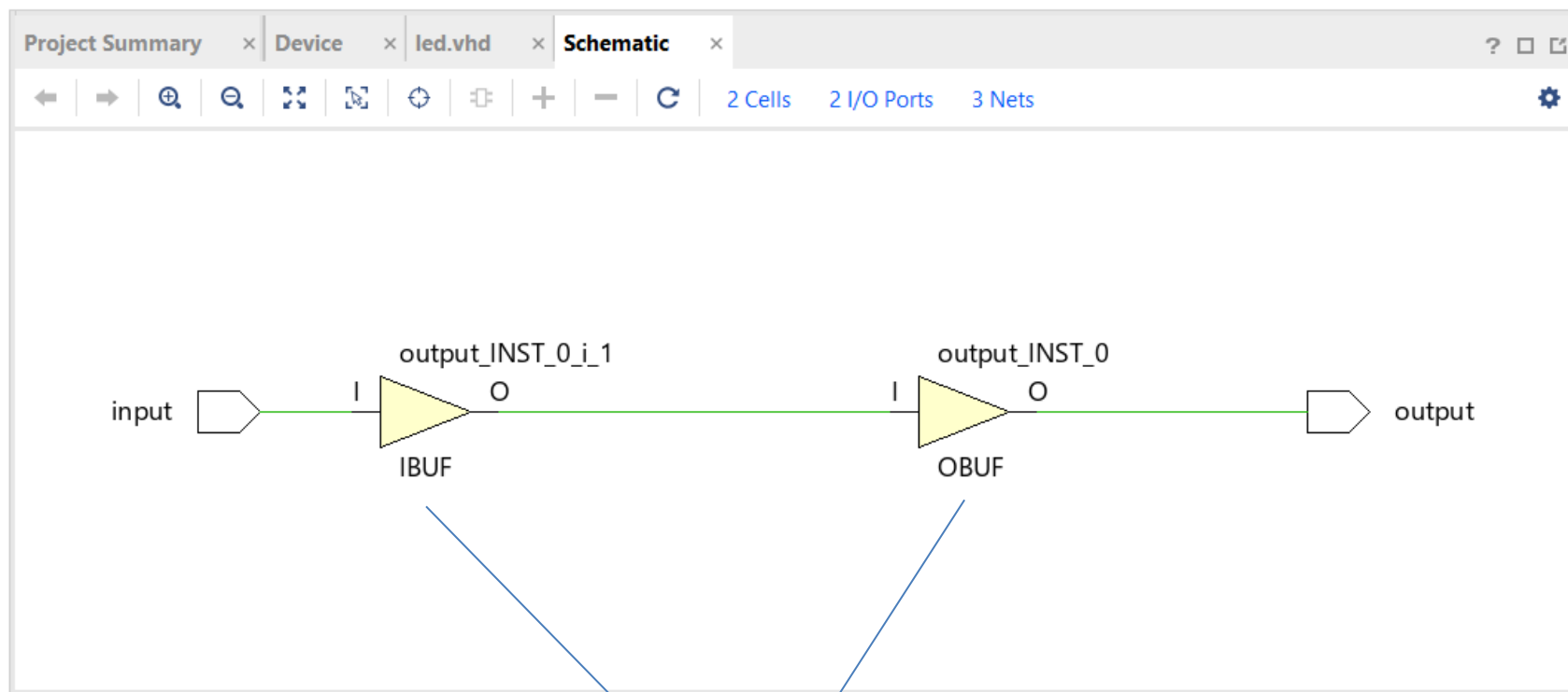
Report DRC

Report Noise

Report Utilization

Report Power

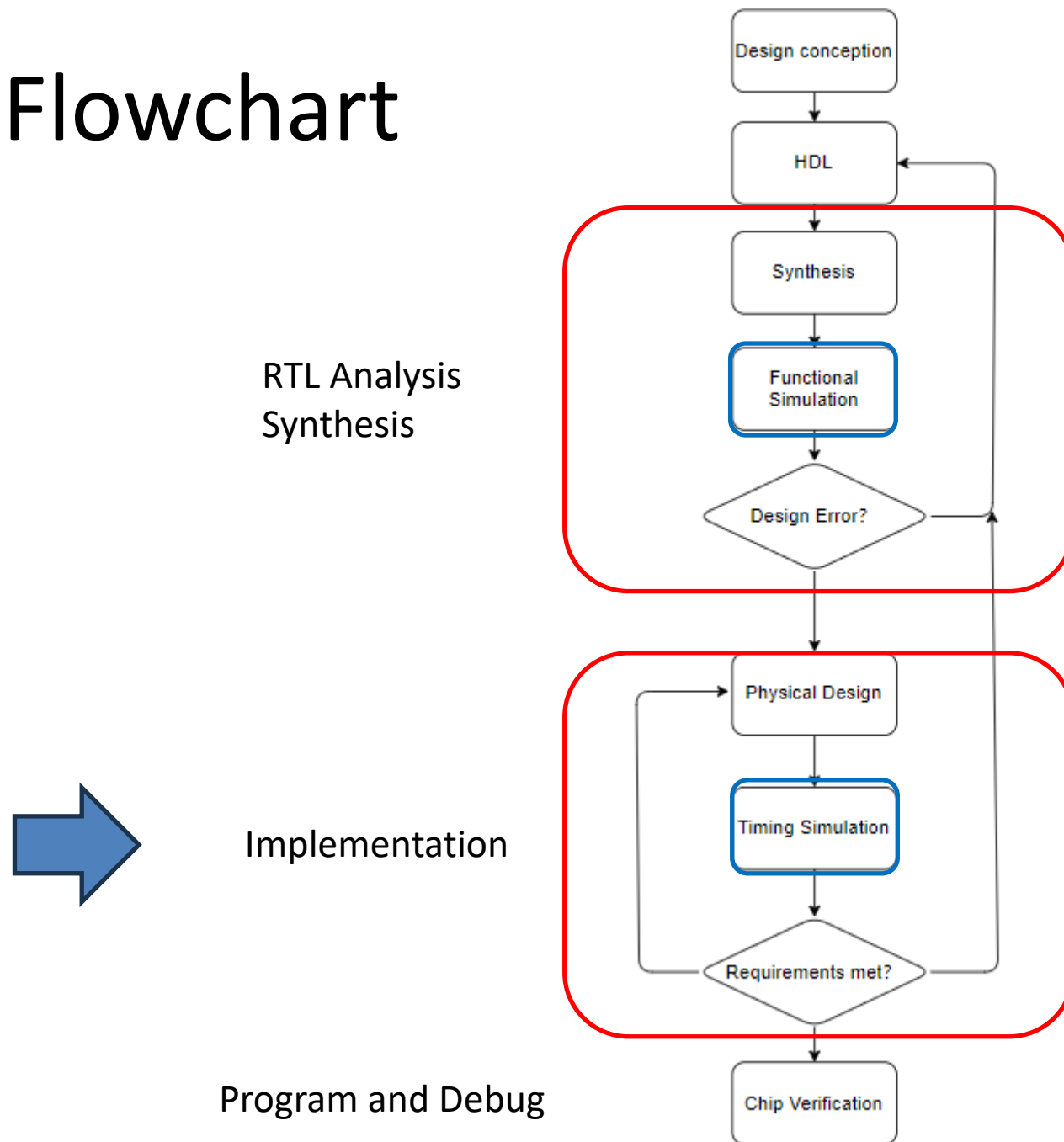
Schematic



FPGA的IO口为电气考量而默认添加的缓冲

与RTL Schematic对比，更底层

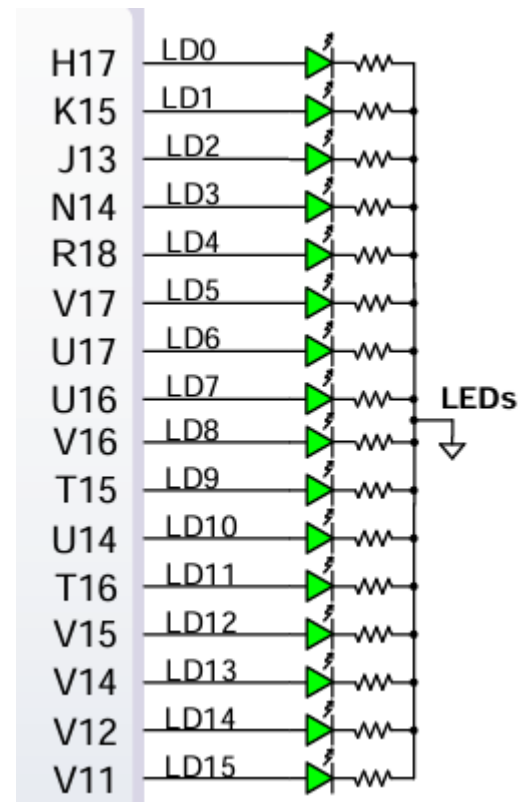
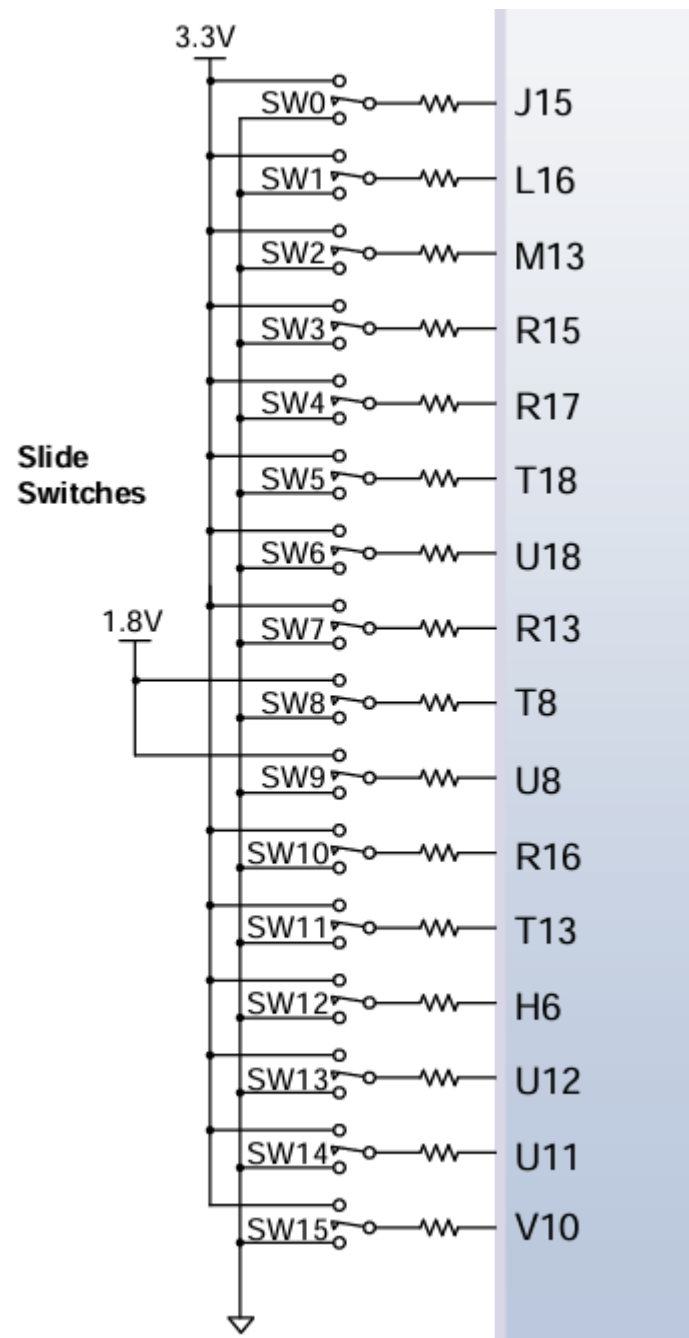
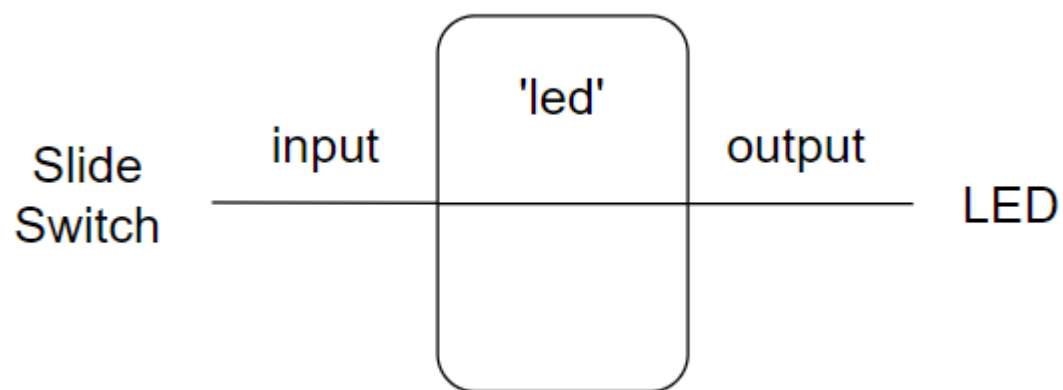
EDA Flowchart



Verification

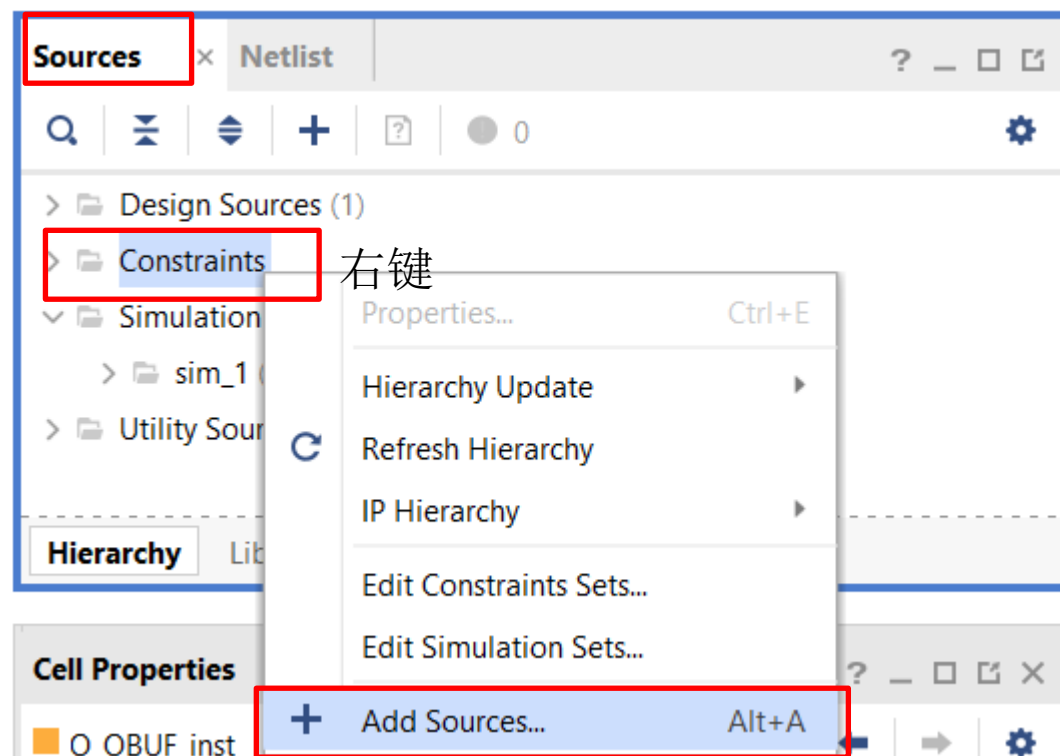
设计验证

使用开发板SW0控制LD0



Create Constran File

约束文件用于指定电平和进行管脚分配等



Create Constrain File



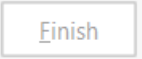
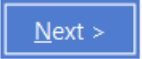
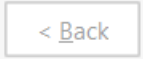
Add Sources

This guides you through the process of adding and creating sources for your project

☒ Add or create constraints

☐ Add or create design sources

☐ Add or create simulation sources



Create Constrain File

Add or Create Constraints

Specify or create constraint files for physical and timing constraint to add to your project.

Specify constraint set: constrs_1 (active)

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Use Add Files or Create File buttons below

Add Files

Create File

☐ Copy constraints files into project

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< Back

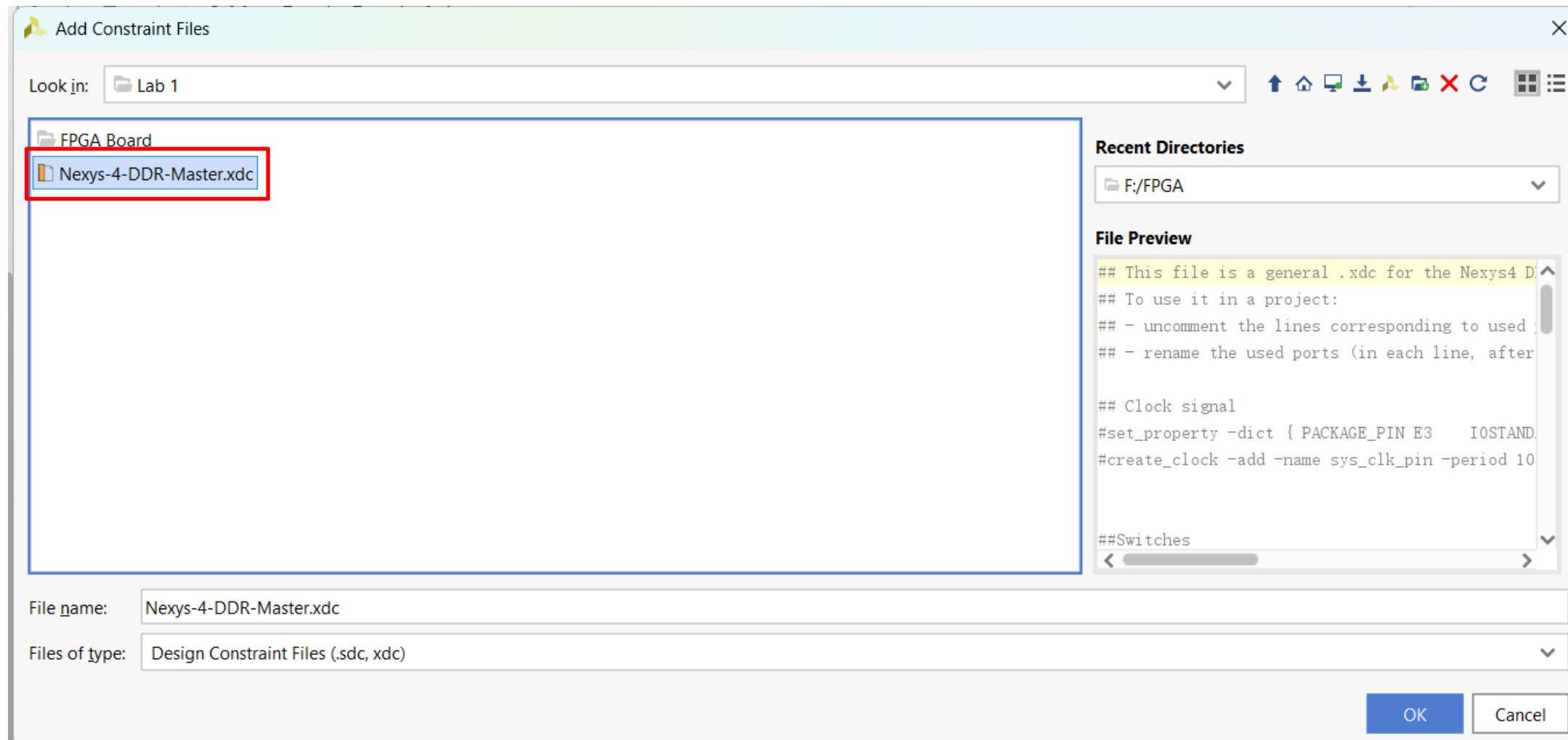
Next >

Finish

Cancel

Create Constran File

从BB上下载板卡提供的约束文件



Create Constrains File

Add or Create Constraints

Specify or create constraint files for physical and timing constraint to add to your project.

Specify constraint set: constrs_1 (active)

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Constraint File	Location
Nexys-4-DDR-Master.xdc	H:\Course\Digital System Design\lab\Lab 1

Add Files

Create File

☒ Copy constraints files into project

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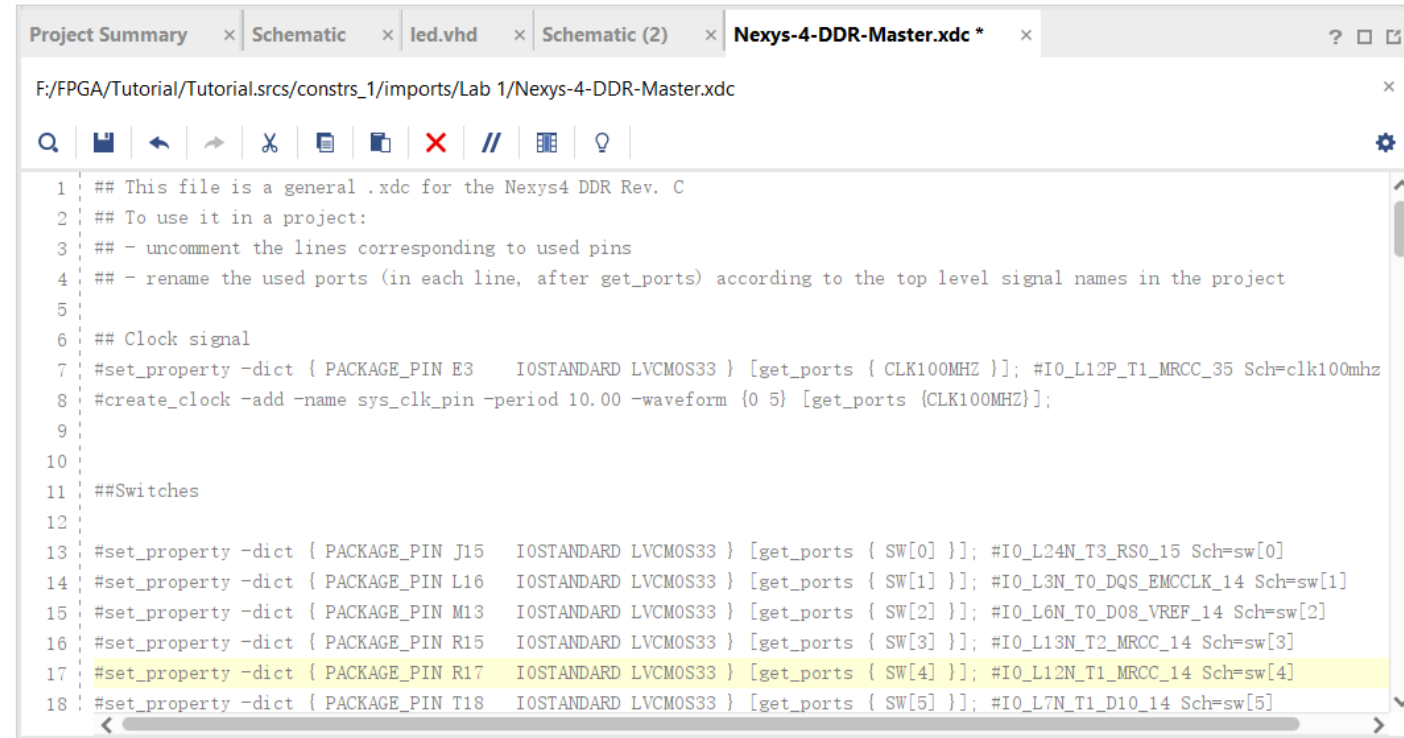
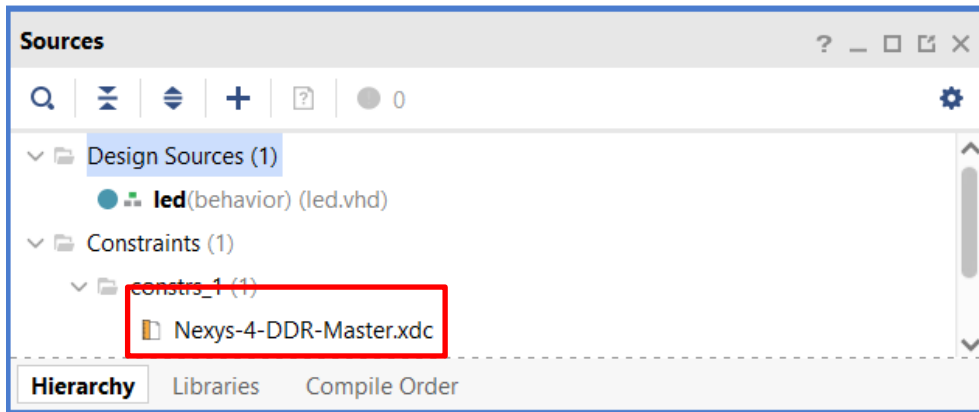
< Back

Next >

Finish

Cancel

Create Constrain File



Create Constrains File

修改约束文件的第13行和第33行

```
##Switches
```

```
set_property -dict { PACKAGE_PIN J15    IOSTANDARD LVCMOS33 } [get_ports { input }]; #IO_L24N_T3_RS0_15 Sch=sw[0]
```

```
## LEDs
```

```
set_property -dict { PACKAGE_PIN H17    IOSTANDARD LVCMOS33 } [get_ports { output }]; #IO_L18P_T2_A24_15 Sch=led[0]
```

#为xdc文件中的注释符号

删除使用引脚所在行的注释符号

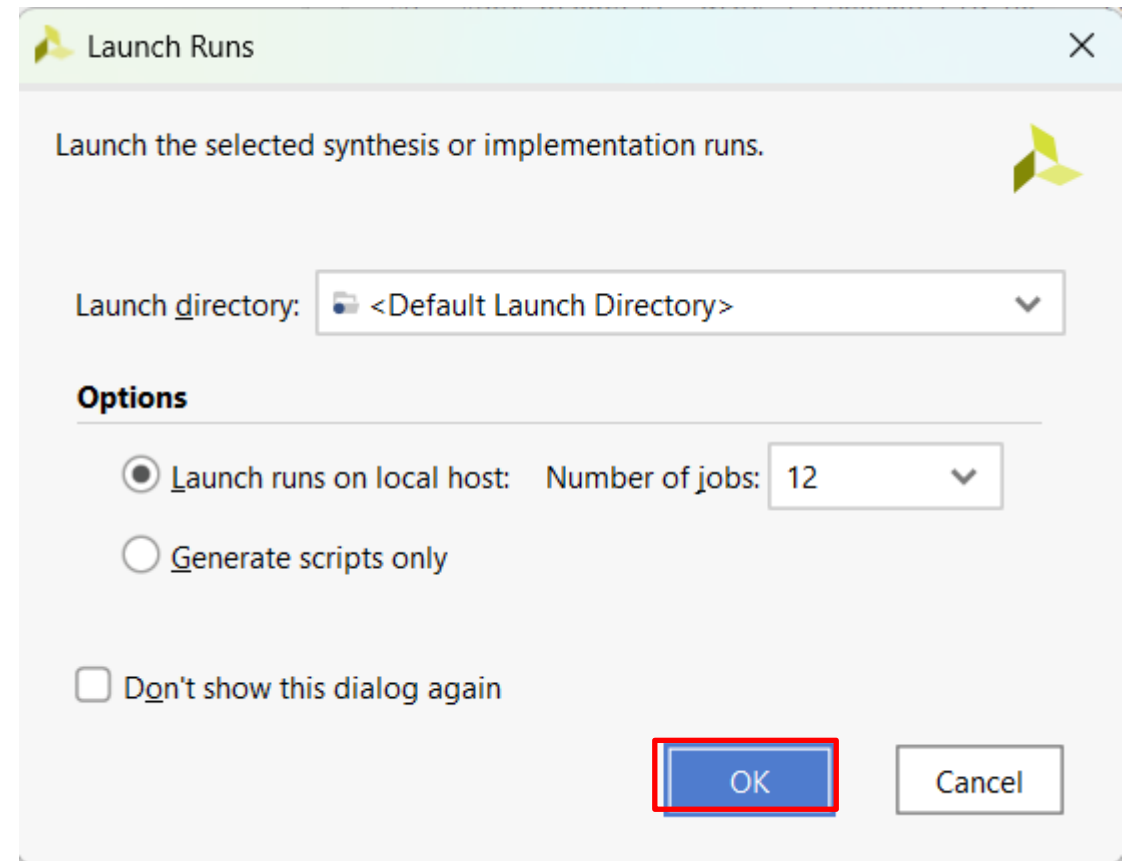
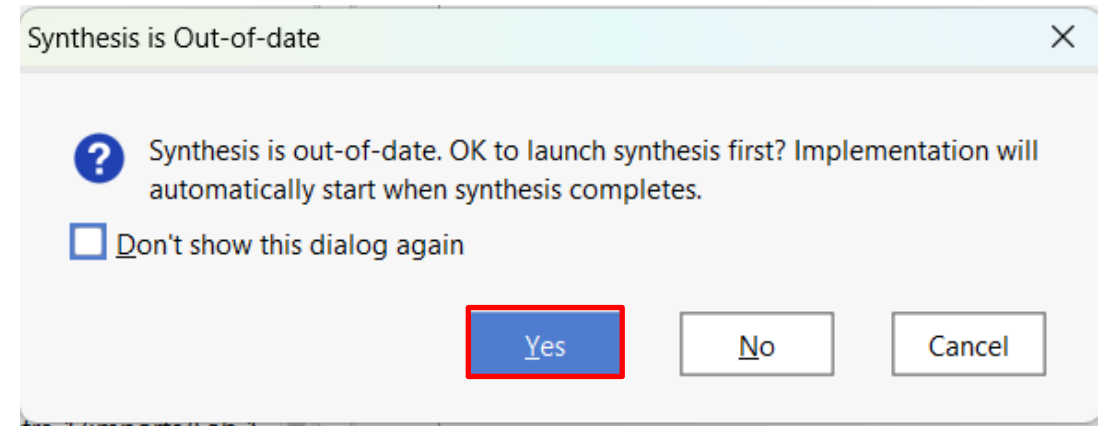
修改默认端口名为设计中的信号名称（每行get_ports 之后）

保存后会出现：

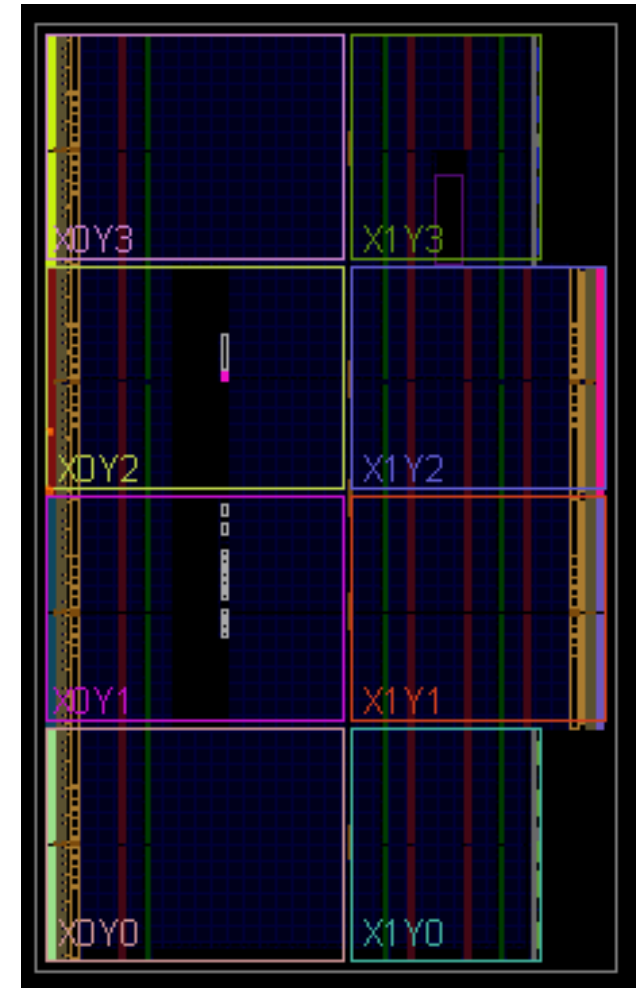
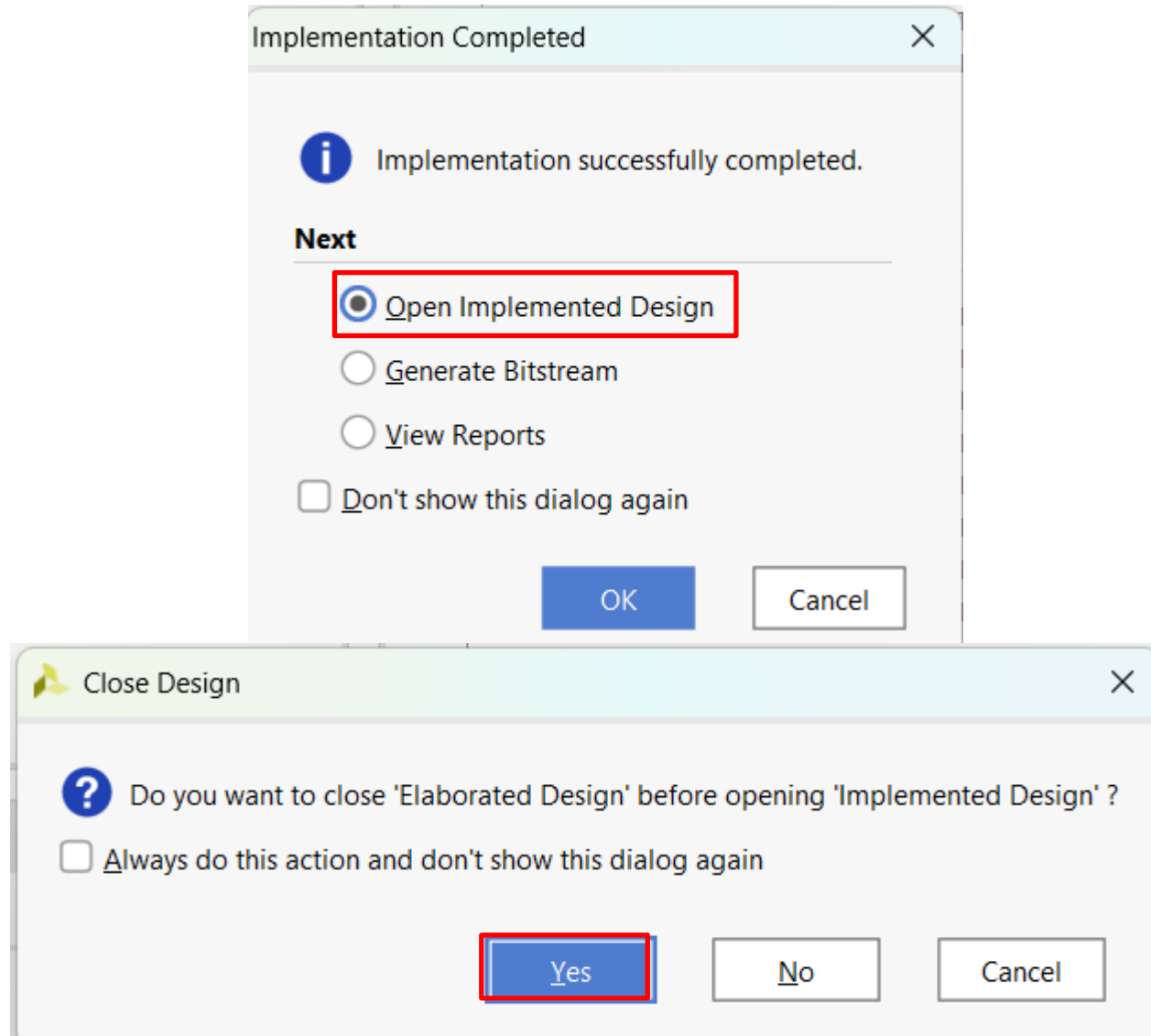
ⓘ Elaborated Design is out-of-date. Constraints were modified. [details](#) [Reload](#)

Implementing the Design

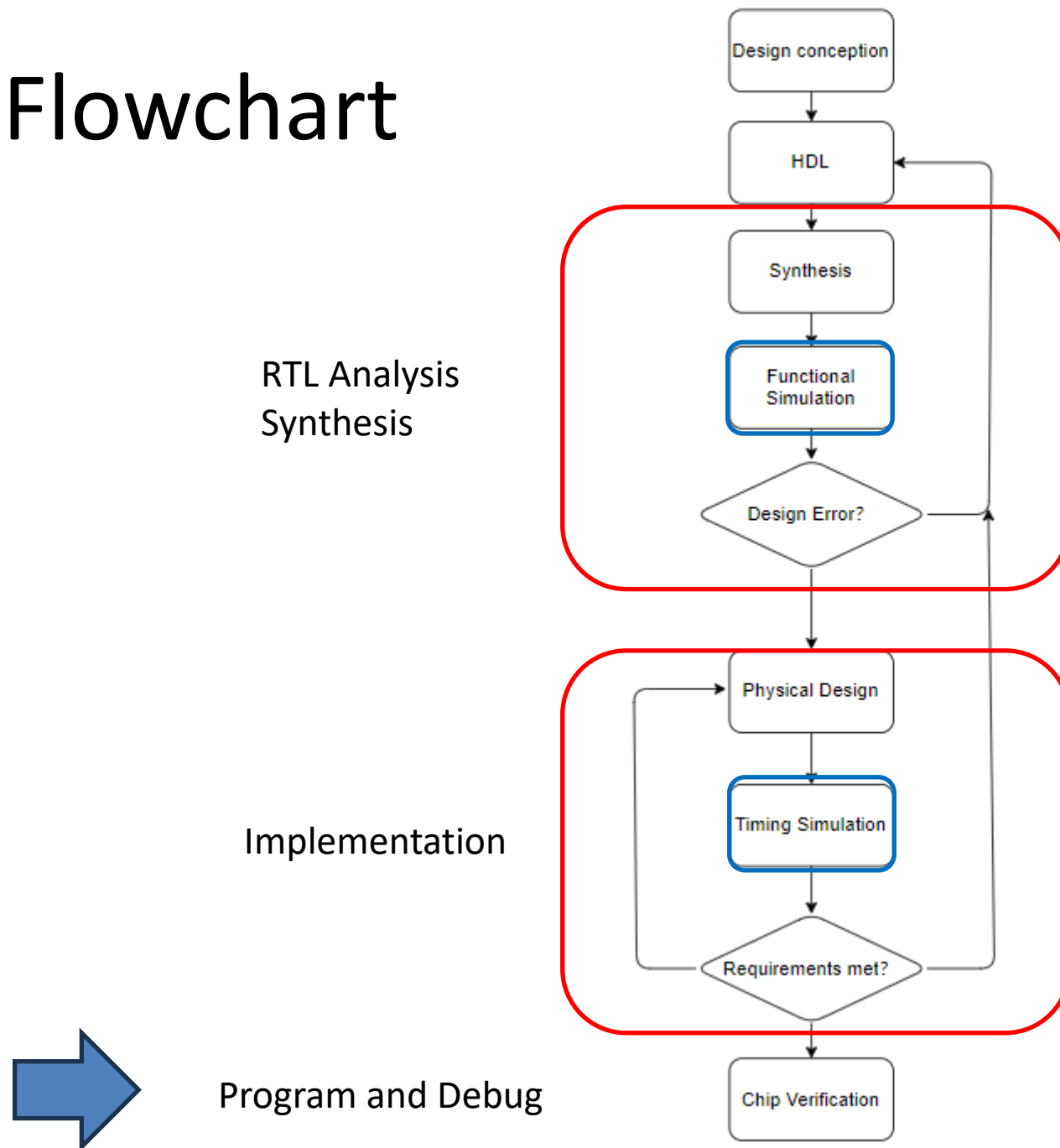
- Report Methodology
 - Report DRC
 - Report Noise
 - Report Utilization
- Report Power
- Schematic
- IMPLEMENTATION
 - Run Implementation**
 - Open Implemented Design
- PROGRAM AND DEBUG
 - Generate Bitstream
 - Open Hardware Manager



Implementing the Design



EDA Flowchart



FPGA Configuration

▼ IMPLEMENTATION

▶ Run Implementation

▼ Open Implemented Design

Constraints Wizard

Edit Timing Constraints

🕒 Report Timing Summary

Report Clock Networks

Report Clock Interaction

📋 Report Methodology

Report DRC

Report Noise

Report Utilization

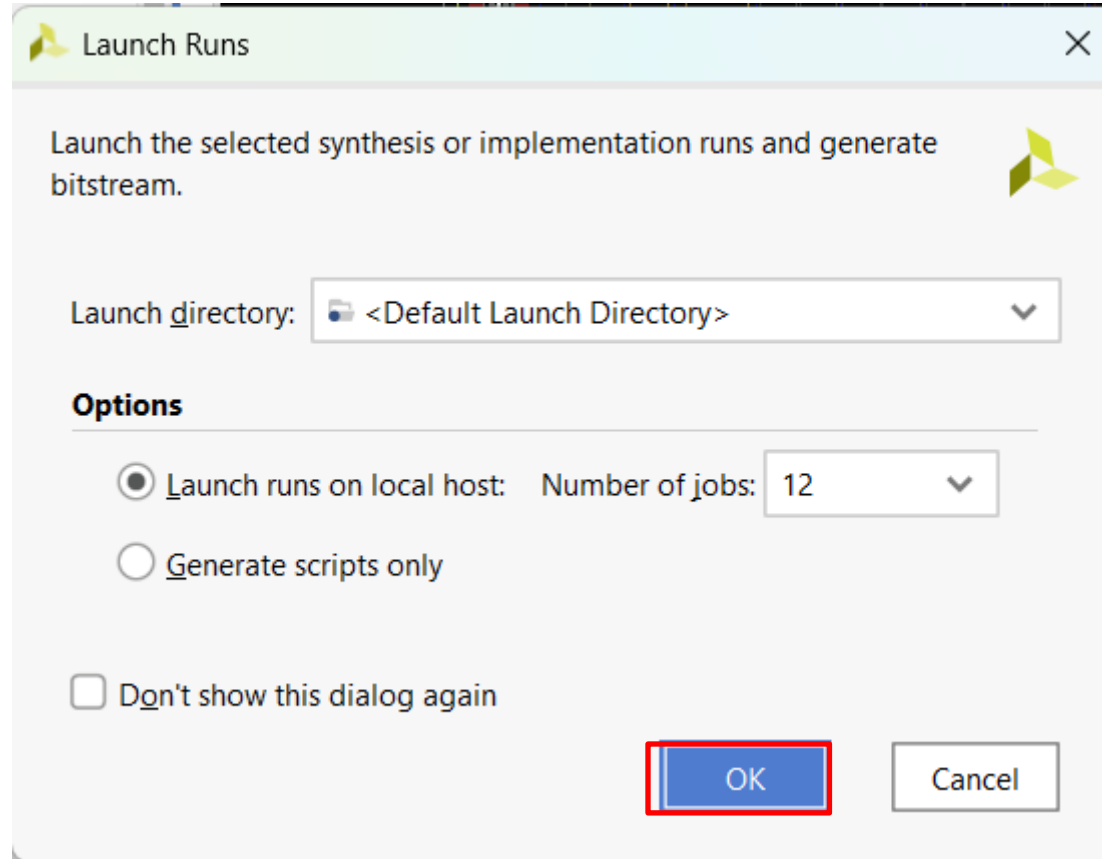
🔌 Report Power

🔧 Schematic

▼ PROGRAM AND DEBUG

⬇️ Generate Bitstream

> Open Hardware Manager



Launch Runs

Launch the selected synthesis or implementation runs and generate bitstream.

Launch directory: <Default Launch Directory>

Options

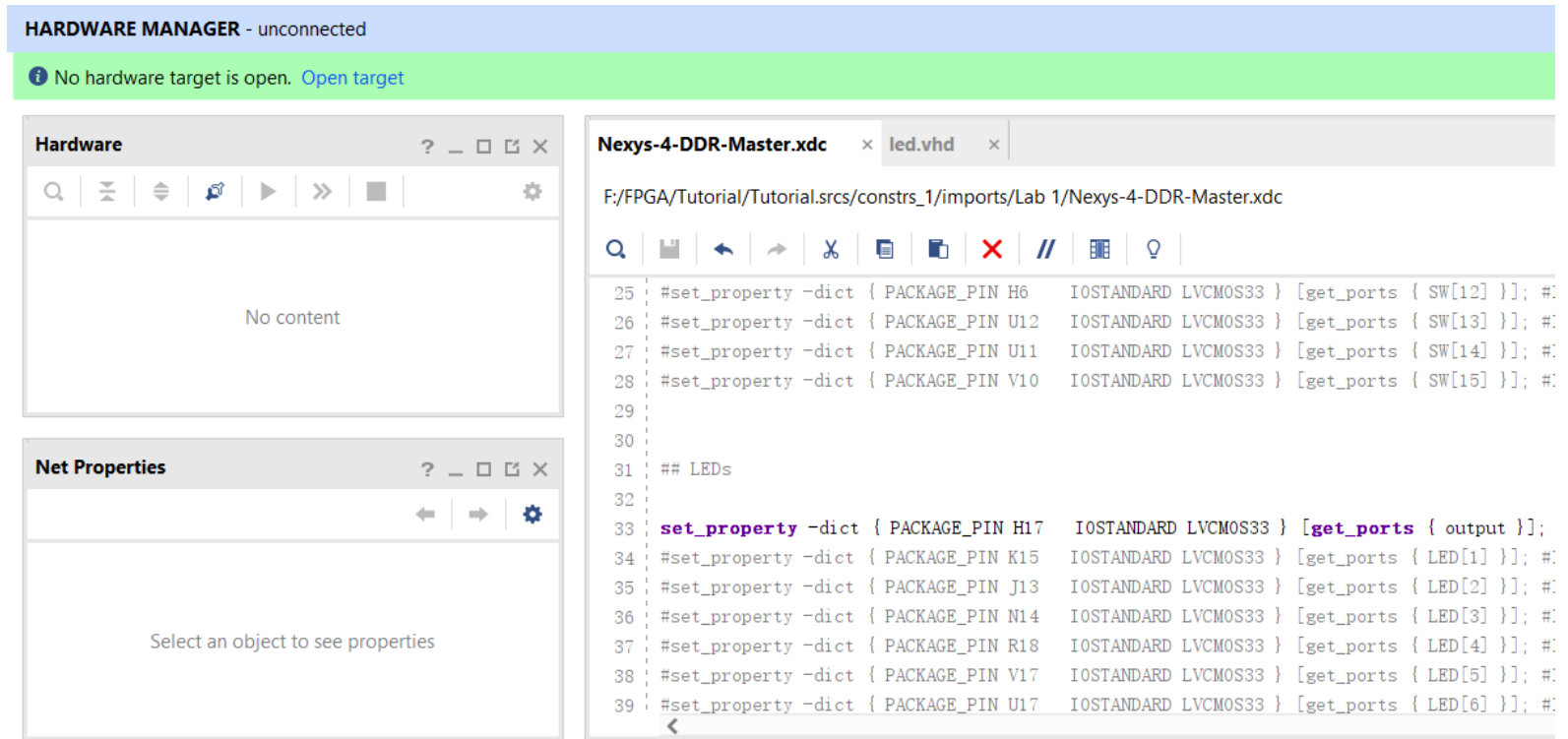
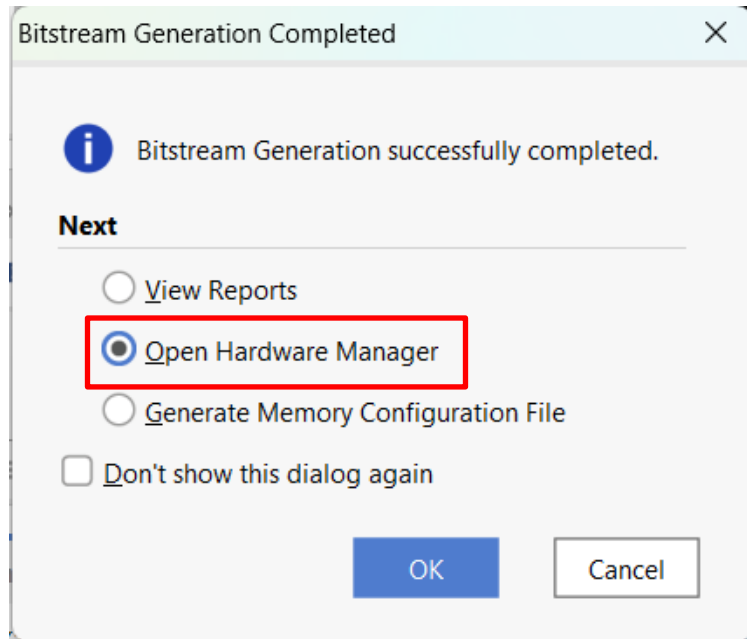
☒ Launch runs on local host: Number of jobs: 12

☐ Generate scripts only

☐ Don't show this dialog again

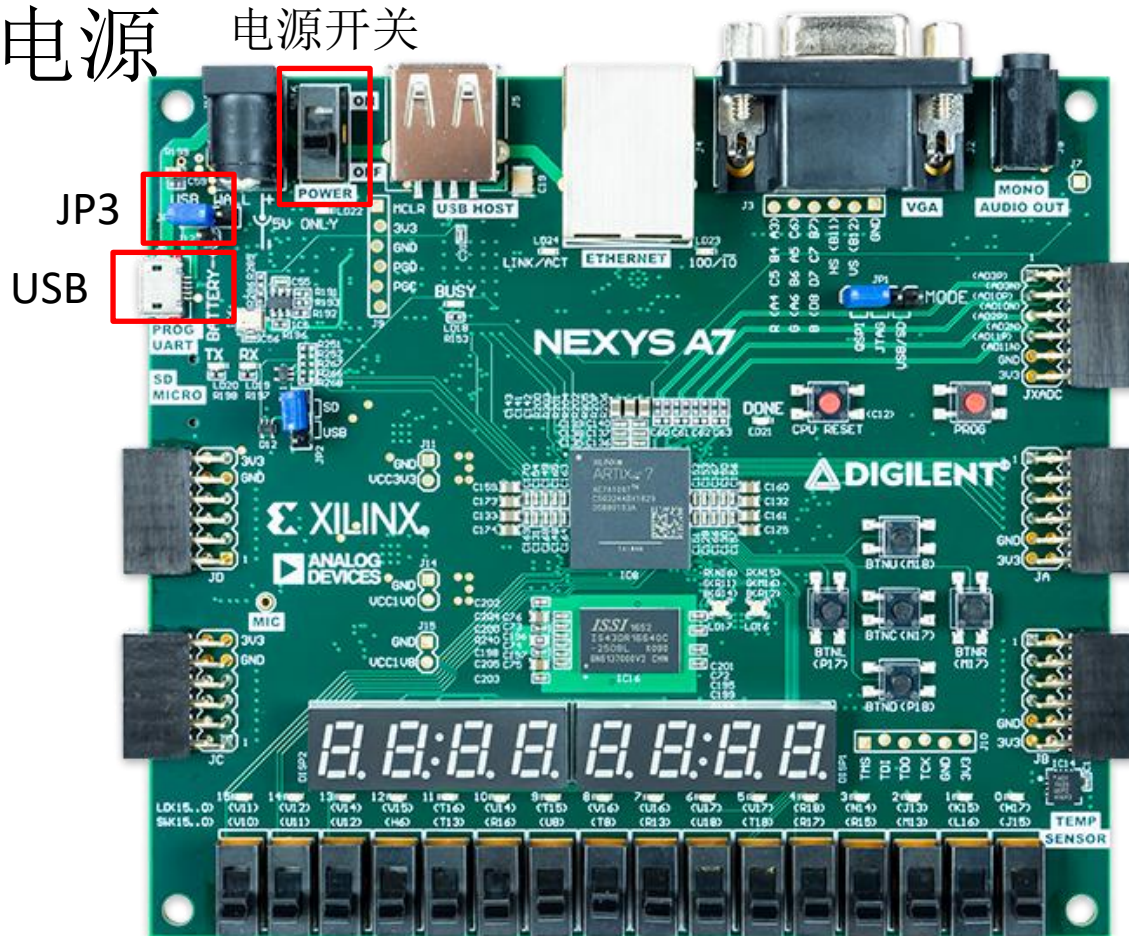
OK Cancel

Implementing the Design

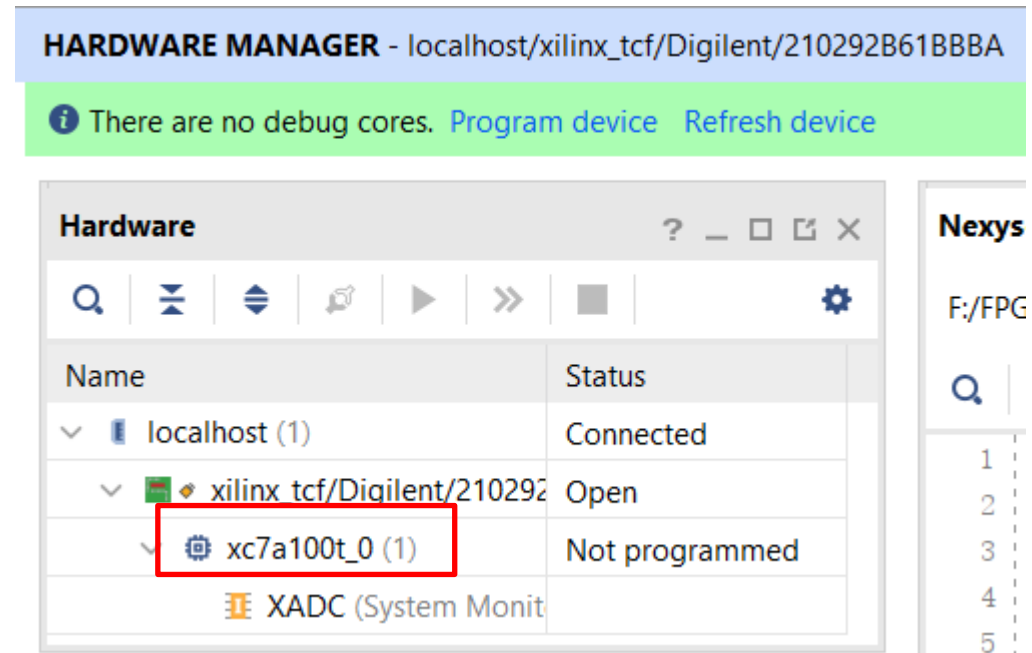
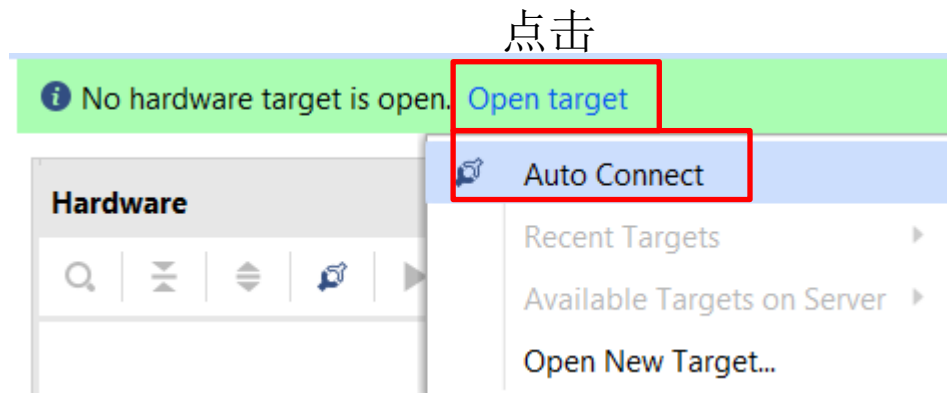


Implementing the Design

将开发板用USB线连接到电脑，请确认JP3设置为从USB供电，之后打开电源



Implementing the Design



Implementing the Design

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210292B61BBBA

There are no debug cores. **Program device** Refresh device

Hardware	
Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210292	Open
xc7a100t_0 (1)	Not programmed
XADC (System Monit	

Nexys

F:/FPG



1
2
3
4
5

Program Device

Select a bitstream programming file and download it to your hardware device. You can optionally select a debug probes file that corresponds to the debug cores contained in the bitstream programming file.

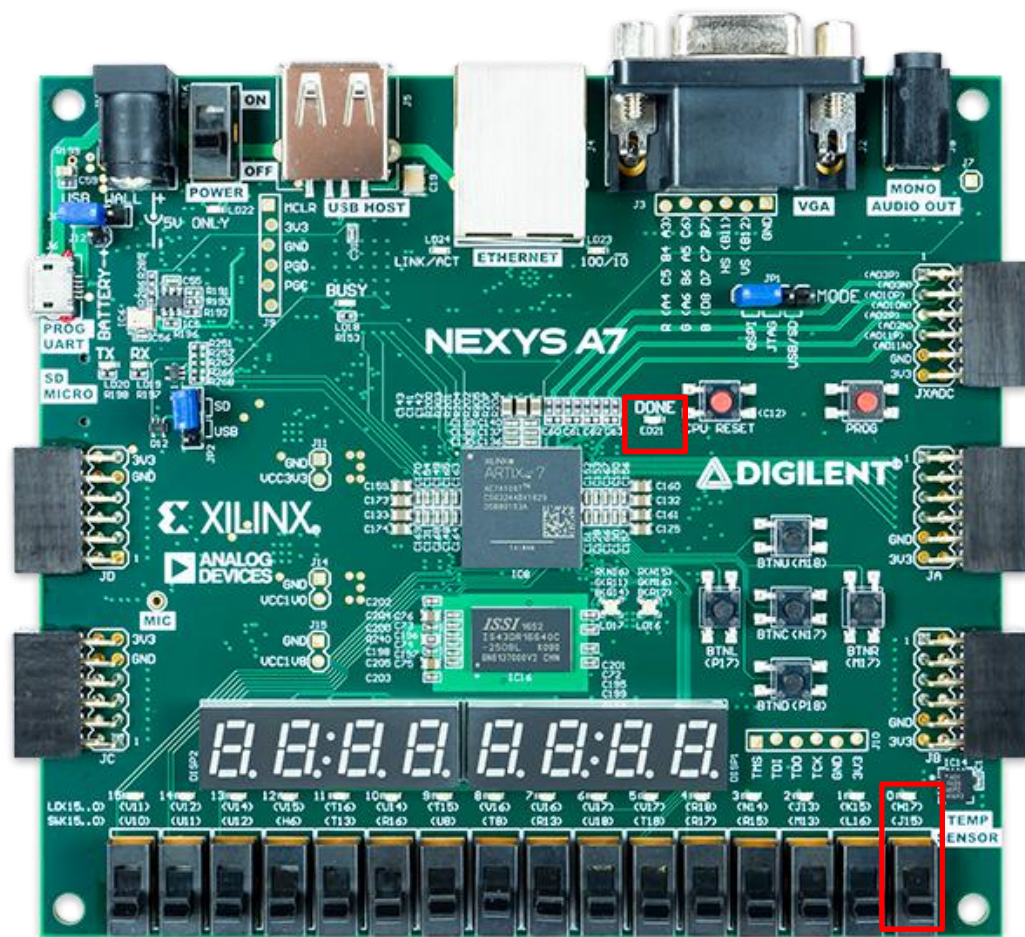
Bitstream file: F:/FPGA/Tutorial/Tutorial.runs/impl_1/led.bit

Debug probes file:

☒ Enable end of startup check

Program Cancel

Verify the Design

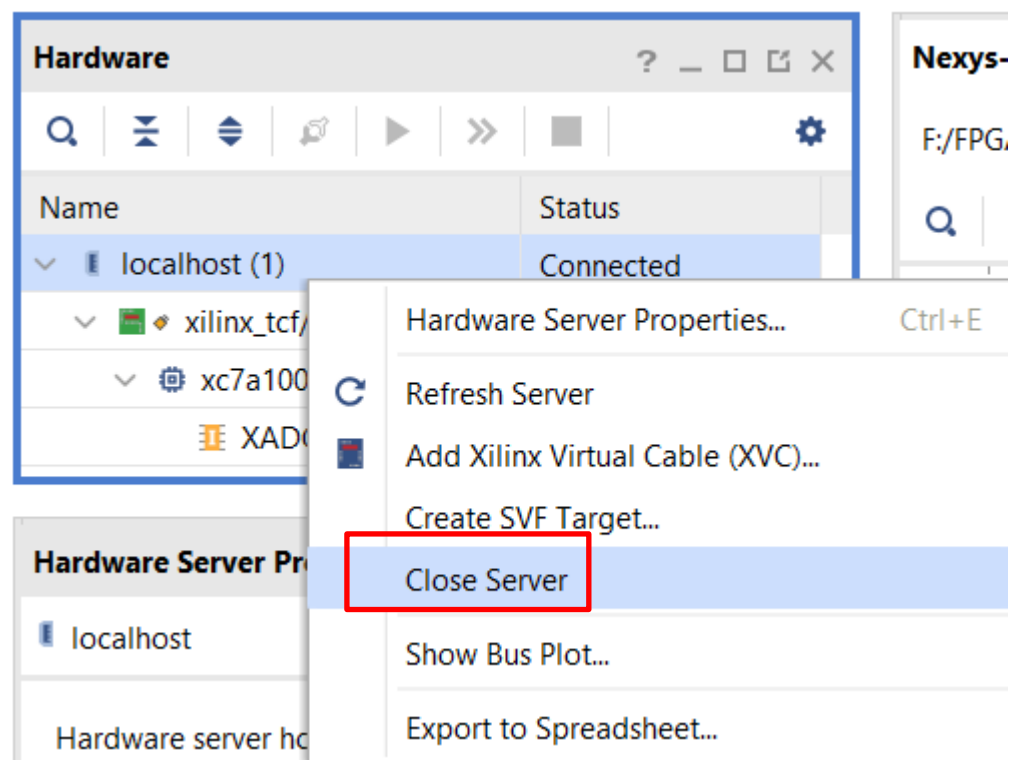


完成灯(DONE,LD21)亮起后就可以进行测试了

拨动最右侧的开关控制LED灯的亮暗

Verify the Design

验证完成后



Programing PROM

FPGA重新启动发现程序会消失
固化到PROM中

The screenshot displays the Xilinx IDE interface. On the left, the 'Flow Navigator' pane shows the 'PROJECT MANAGER' section with 'Settings' highlighted by a red box. Below it are 'Add Sources', 'Language Templates', and 'IP Catalog'. The 'IP INTEGRATOR' section includes 'Create Block Design', 'Open Block Design', and 'Generate Block Design'. The main area on the right is the 'HARDWARE MANAGER' for the device 'localhost/xilinx_tcf/Digilent/210292B61BBBA'. It features a green status bar with the message 'There are no debug cores. Program device Refresh device'. Below this is a 'Hardware' table with columns 'Name' and 'Status'. The table lists three items: 'localhost (1)' with status 'Connected', 'xilinx_tcf/Digilent/210292B61BBBA (1)' with status 'Open', and 'xc7a100t_0 (1)' with status 'Programmed'. A 'Properties' pane is visible at the bottom right. On the far right, a 'Nexys' panel shows 'F:/FPG' and a list of numbers 1 through 9.

Flow Navigator

- PROJECT MANAGER
 - Settings
 - Add Sources
 - Language Templates
 - IP Catalog
- IP INTEGRATOR
 - Create Block Design
 - Open Block Design
 - Generate Block Design

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210292B61BBBA

There are no debug cores. [Program device](#) [Refresh device](#)

Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210292B61BBBA (1)	Open
xc7a100t_0 (1)	Programmed
XADC (System Monitor)	

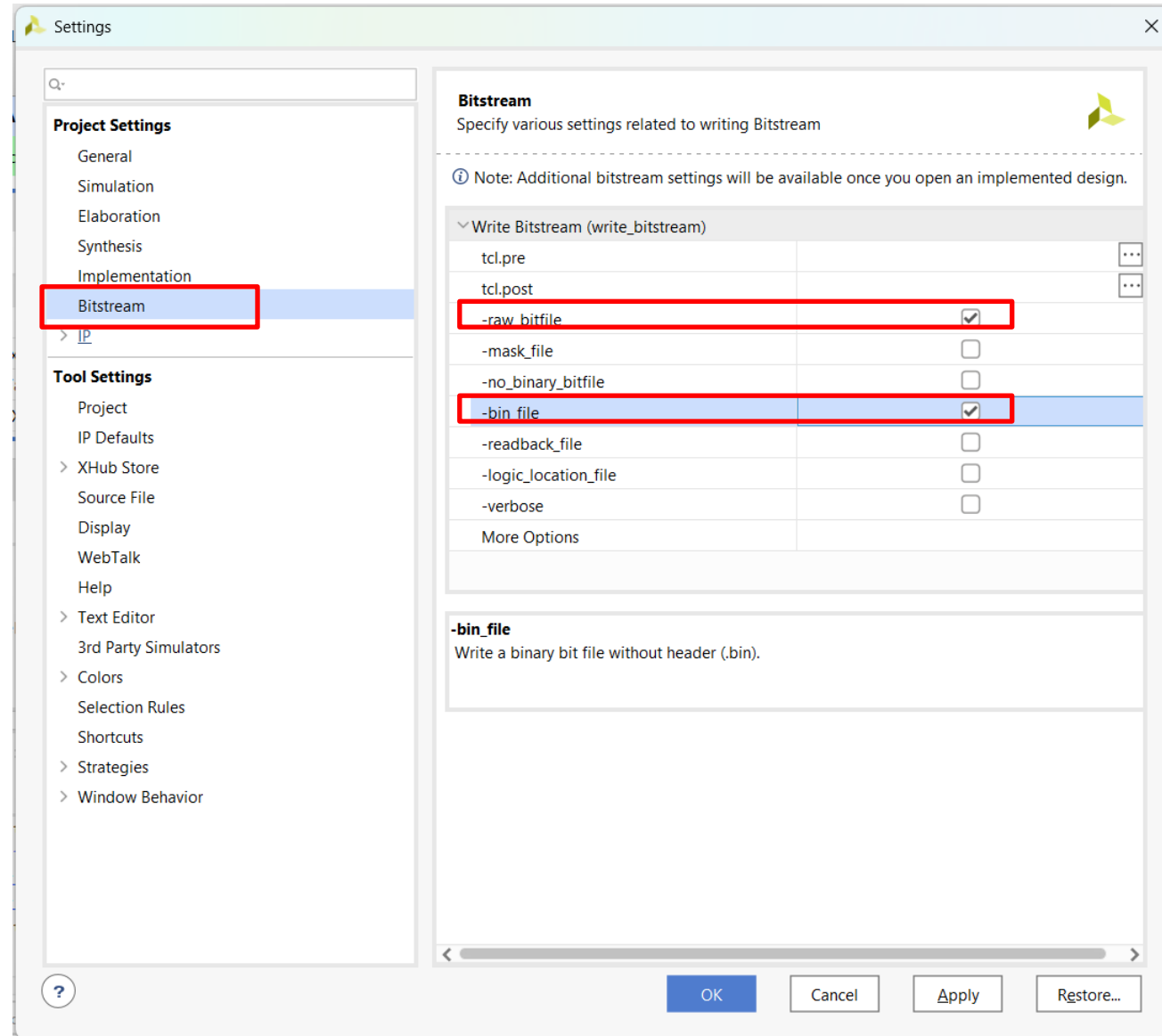
Properties

Nexys

F:/FPG

1
2
3
4
5
6
7
8
9

Programing PROM



Programing PROM

The screenshot displays the Xilinx IDE interface, specifically the **HARDWARE MANAGER** window. The title bar indicates the connection path: **HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210292B61BBBA**.

A green status bar at the top of the Hardware Manager contains the message: "There are no debug cores. [Program device](#) [Refresh device](#)". The **Refresh device** button is highlighted with a red rectangle.

The main area of the Hardware Manager is divided into two panes:

- Hardware**: A table listing the hardware components connected to the system.
- Properties**: A pane for viewing the properties of the selected hardware component.

The **Hardware** table has the following data:

Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210292B61BBBA	Open
xc7a100t_0 (1)	Programmed
XADC (System Monitor)	

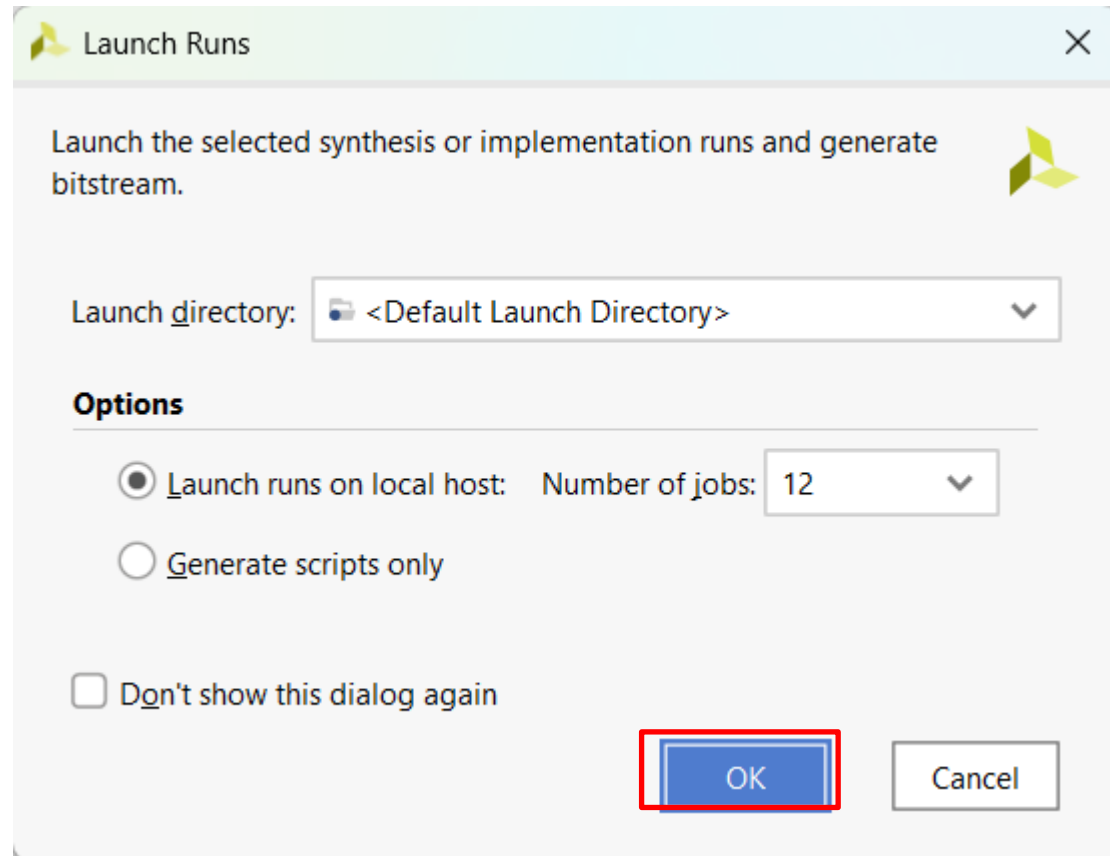
The **Properties** pane is currently empty.

On the left side of the IDE, the **Flow Navigator** is visible, showing the **PROJECT MANAGER** and **IP INTEGRATOR** sections. The **PROJECT MANAGER** section includes options like **Settings**, **Add Sources**, **Language Templates**, and **IP Catalog**. The **IP INTEGRATOR** section includes **Create Block Design**, **Open Block Design**, and **Generate Block Design**.

On the right side, the **Nexys** board is selected, showing the **F:/FPG** directory and a list of files numbered 1 through 9.

Programing PROM

- Report Timing Summary
 - Report Clock Networks
 - Report Clock Interaction
- Report Methodology
 - Report DRC
 - Report Utilization
- Report Power
- Schematic
- PROGRAM AND DEBUG**
 - Generate Bitstream**
 - Open Hardware Manager**
 - Open Target
 - Program Device
 - Add Configuration Memory



Programing PROM

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210292B61B8BA

There are no debug cores. [Program device](#) [Refresh device](#)

Hardware

Name	Status
localhost (1)	Connected
xilinx_tcf/Digilent/210292B61B8BA	Open
xc7a100t_0 (1)	

Hardware Device Properties

xc7a100t_0

xc7a100t

General Properties

Tcl Console Messages Report

Nexys-4-DDR-Master.xdc

F:/FPGA/Tutorial/Tutorial.srcs/c

1 ## This file is a ge
2 ## To use it in a pr

Hardware Device Properties... Ctrl+E

Program Device...

Verify Device...

Refresh Device

Show Bus Plot...

Add Configuration Memory Device...

Boot from Configuration Memory Device

Program BBR Key...

Clear BBR Key...

Program eFUSE Registers...

Export to Spreadsheet...

Add Configuration Memory Device

Choose a configuration memory part.

Device: xc7a100t_0

Filter

Manufacturer: All Type: spi Density (Mb): 128 Width: All

[Reset All Filters](#)

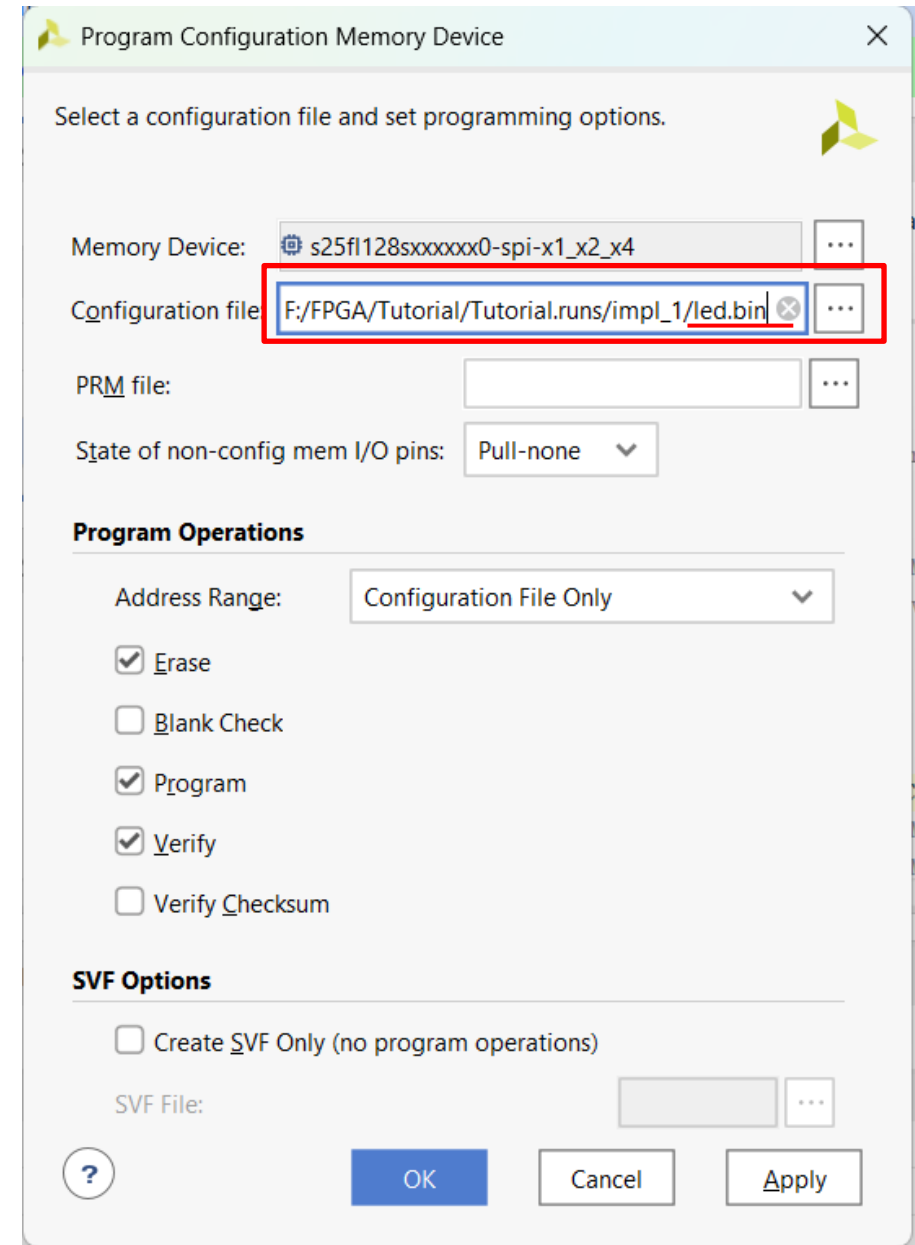
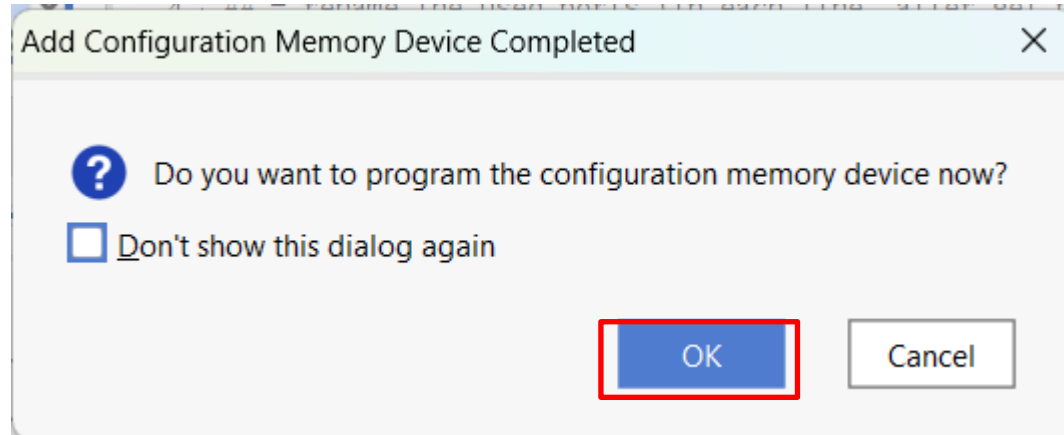
Select Configuration Memory Part

Search: s25fl128 (3 matches)

Name	Part	Manufact...	Alias	Family	Type	Density (...)	Wic
s25fl128l-spi-x1_x2_x4	s25fl128l	Spansion		s25flxxxl	spi	128	x1_
s25fl128sxxxxx0-spi-x1_x2_x4	s25fl128sxxxxx0	Spansion	s25fl127s-spi-x1_x2_x4	s25flxxxs	spi	128	x1_
s25fl128sxxxxx1-spi-x1_x2_x4	s25fl128sxxxxx1	Spansion		s25flxxxs	spi	128	x1_

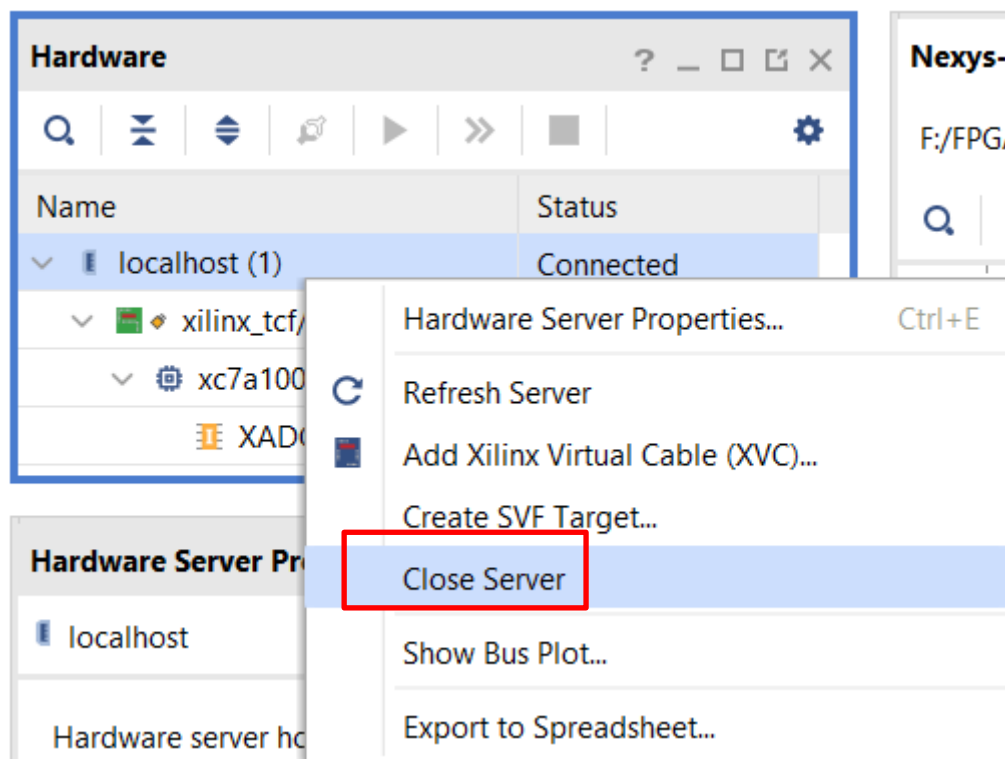
[?](#) [OK](#) [Cancel](#)

Programing PROM



Programing PROM

烧录完成之后，要断开连接后才会开始工作



Exercise

如果用SW3~SW0分别控制LD3~LD0，应如何进行设计？

